

Elliptic PDE notes

Shreeram Jaikisan

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I am grateful to the authors whose books and notes have helped me learn elliptic partial differential equations. During my studies, I have often come across explanations and notes written by others that helped me understand things that I could not quite understand from textbooks alone.

I have found these kinds of resources very valuable, and they have influenced how I have approached this subject and the way I have written these notes. I hope that, in some small way, these notes can be useful to someone else trying to learn the subject.

These notes are also a work in progress. As I study more and come across new ideas, better explanations, and different points of view, I expect to add to and change them. Some parts may therefore look different as my understanding of the subject develops.

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Chapter 1

Calculus of Variations

So far, we have seen the theory of Sobolev spaces and how it is useful in solving linear elliptic equations. But what about nonlinear elliptic equations? How do we solve them? Well, there is no single general theory for solving equations of this type. One approach, however, is provided by the Calculus of Variations. Suppose we have a nonlinear operator $A[\cdot]$, such that

$$A[u] = 0, \tag{1.1}$$

If we assume $A[\cdot]$ is the derivative of an appropriate energy functional $I[\cdot]$, then we can write

$$A[\cdot] = I'[\cdot]$$

So we just have to solve for

$$I'[u] = 0$$

This is similar to what we do in multivariable calculus, where we look for the critical points of a function. In the same way, here we are looking for the critical points of the energy functional $I[\cdot]$. In particular, if $I[\cdot]$ attains a minimum at some function u , then u is a critical point and hence a solution of our PDE [1.1](#).

In the next few sections we will deal with just that.

1.1 First variation, the Euler Lagrange Equation

Notation:

Let U be bounded, open subset of $\mathbb{R}^n$, with smooth boundary. Define a smooth

function

$$L : \mathbb{R}^n \times \mathbb{R} \times \bar{U} \rightarrow \mathbb{R}$$

the arguments of L , i.e. $L(p, z, x)$,

we replace p with $Dw(x)$, and z with $w(x)$.

We also set

$$\begin{cases} D_p L = (L_{p_1}, \dots, L_{p_n}) \\ D_z L = L_z \\ D_x L = (L_{x_1}, \dots, L_{x_n}) \end{cases}$$

We now assume our functional $I[\cdot]$ to have the form

$$I[w] = \int_U L(Dw(x), w(x), x) dx \quad (1.2)$$

for smooth function $w : \bar{U} \rightarrow \mathbb{R}$ satisfying the boundary condition

$$w = g, \quad \text{on } \partial U \quad (1.3)$$

Now, suppose we have a smooth function u , satisfying the boundary condition 1.3, that minimizes functional $I[\cdot]$. We will see that u solves a nonlinear differential equation.

To see this, choose a test function $v \in C_0^\infty(U)$, and define the real valued function

$$i(\tau) = I[u + \tau v] \quad (1.4)$$

on the boundary, ∂U , we have $u = g = u + \tau v$, and we have assumed u is a minimizer of $I[\cdot]$. So, function i has a minima at the point $\tau = 0$, and

$$\begin{aligned} i'(0) &= 0 = I'[u + \tau v] \\ \implies i'(\tau) &\stackrel{\text{chain rule}}{=} \int_U \sum_{i=1}^n L_{p_i}(Du + \tau Dv, u + \tau v, x) v_{x_i} \end{aligned} \quad (1.5)$$

$$+ L_z(Du + \tau Dv, u + \tau v, x) v dx \quad (1.6)$$

$$(1.7)$$

At $\tau = 0$,

$$\begin{aligned} i'(0) = 0 &= \int_U \sum_{i=1}^n L_{p_i}(Du, u, x) v_{x_i} + L_z(Du, u, x) v \, dx \\ \implies &\stackrel{\substack{v \text{ has compact support} \\ \text{int by parts}}}{=} \int_U \left[- \sum_{i=1}^n (L_{p_i}(Du, u, x))_{x_i} + L_z(Du, u, x) \right] v \, dx = 0 \end{aligned}$$

This implies, u solves the quasi linear second order PDE

$$\sum_{i=1}^n (L_{p_i}(Du, u, x))_{x_i} + L_z(Du, u, x) = 0, \quad \text{in } u \quad (1.8)$$

Equation 1.8 is the Euler-Lagrange equation associated with the energy functional $I[\cdot]$ define by 1.2

Remark 1.1.1. Any smooth minimizer of $I[\cdot]$ is a solution of Euler-Lagrange equation 1.8.

1.2 Second variation

Since $\tau = 0$ is the minimum of the real-valued function i , we must have

$$i''(0) \geq 0$$

Differentiating equation 1.7 again, we get

$$\begin{aligned} i''(\tau) &= \int_U \sum_{i,j=1}^n L_{p_i p_j}(Du + \tau Dv, u + \tau v, x) v_{x_i} v_{x_j} \\ &\quad + 2 \sum_{i=1}^n L_{p_i z}(Du + \tau Dv, u + \tau v, x) v_{x_i} v \\ &\quad + L_{zz}(Du + \tau Dv, u + \tau v, x) v^2 \, dx \end{aligned}$$

setting $\tau = 0$, we get

$$\begin{aligned} 0 \leq i'(0) &= \int_U \sum_{i,j=1}^n L_{p_i p_j}(Du, u, x) v_{x_i} v_{x_j} \\ &+ 2 \sum_{i=1}^n L_{p_i z}(Du, u, x) v_{x_i} v + L_{zz}(Du, u, x) v^2 dx \end{aligned} \quad (1.9)$$

The above inequality holds for every the test functions $v \in C_0^\infty(U)$.

$$\begin{aligned} v &\in C_0^\infty(U) \\ &\xRightarrow{\text{density theorem}} v \in W_0^{1,\infty}(U) \end{aligned}$$

$\implies$ By the Sobolev embedding theorem, every $v \in W^{1,\infty}(U)$ is Lipschitz.

$\implies$ we can take v to be Lipschitz continuous with zero trace.

Fix $\xi \in \mathbb{R}^n$ and define

$$v(x) = \epsilon \rho\left(\frac{x \cdot \xi}{\epsilon}\right) \zeta(x), \quad x \in U \quad (1.10)$$

where $\zeta \in C_0^\infty(U)$ and $\rho : \mathbb{R} \rightarrow \mathbb{R}$ is a periodic zig-zag function defined by

$$\rho(x) = \begin{cases} x, & 0 \leq x \leq \frac{1}{2} \\ 1 - x, & \frac{1}{2} \leq x \leq 1 \end{cases} \quad \rho(x+1) = \rho(x), \quad x \in \mathbb{R}$$

Thus

$$|\rho'| = 1 \text{ a.e.} \quad (1.11)$$

By chain rule, we observe that

$$v_{x_i} = \rho'\left(\frac{x \cdot \xi}{\epsilon}\right) \xi_i \zeta + O(\epsilon) \text{ as } \epsilon \rightarrow 0$$

substituting 1.10 into 1.9, we get

$$0 \leq \int_U \sum_{i,j=1}^n L_{p_i p_j}(Du, u, x) (\rho')^2 \xi_i \xi_j \zeta^2 dx + O(\epsilon)$$

Using 1.11 and letting $\epsilon \rightarrow 0$, we obtain the inequality

$$0 \leq \int_U \sum_{i,j=1}^n L_{p_i p_j}(Du, u, x) \xi_i \xi_j \zeta^2 dx$$

This inequality holds for all $\zeta \in C_0^\infty(U)$, we deduce that

$$\sum_{i,j=1}^n L_{p_i p_j}(Du, u, x) \xi_i \xi_j \geq 0 \quad \xi \in \mathbb{R}^n, x \in U \quad (1.12)$$

1.3 Existence and Uniqueness of Minimizers

In this section, we would be checking under what conditions, minimizers exist. But first of all, why are we interested in minimizers at all? Because in a lot of physical phenomena, low energy state is preferred by the most systems, and low energy means stable.

Our functional $I[\cdot]$. call it f for analogy, like many other smooth functions, could be bounded below, but sometimes, it could not attain infimum. Take e^x for an example. It attains infimum at $-\infty$, which is not a real point. To avoid this situation, we would like to assume that functional $I[\cdot]$ grows as rapidly as $u \rightarrow \infty$. That's where the concept of coercivity comes in.

Assume

$$1 < q < \infty \text{ is fixed} \quad (1.13)$$

Suppose that

$$\begin{cases} \text{there exists constants } \alpha > 0, \beta \geq 0 \text{ such that} \\ L(p, z, x) \geq \alpha |p|^q - \beta \\ \text{for all } p \in \mathbb{R}^n, z \in \mathbb{R}, x \in U \end{cases} \quad (1.14)$$

Therefore,

$$I[w] \geq \delta \|Dw\|_{q,U}^q - \gamma \quad (1.15)$$

for $\gamma = \beta|U|$ and some constant $\delta > 0$.

Therefore, $I[w] \rightarrow \infty$ as $\|Dw\|_q \rightarrow \infty$. condition 1.15 is called *coercivity condition* on $I[\cdot]$.

So far, we have been discussing about smooth functions. But we can extend this to the Sobolev spaces. We define the admissible set as the class of functions

$$\mathcal{A} = \{w \in W^{1,q}(U) | w = g \text{ on } \partial U \text{ in the trace sense}\}$$

Now observe that, in general, a smooth function $f : \mathbb{R} \rightarrow \mathbb{R}$ satisfying a coercivity condition does attain its infimum. To see this, take a sequence $\{x_k\}_{k=1}^\infty$ such that $\{f(x_k)\} \rightarrow m$. Since f is coercive, x_k is bounded. In finite dimension, every bounded sequence has a convergent subsequence. Therefore, $x_{k_j} \rightarrow x_0$. And since, f is smooth, $f(x_{k_j}) \rightarrow f(x_0)$. But we know $f(x_{k_j}) \rightarrow m$. Therefore, $f(x_0) = m$. Hence, f does attain its infimum.

But, now we are dealing with infinite dimensions, and $I[\cdot]$ might not always be continuous. Our aim is to show that there exists a subsequence of minimizing function, since normal sequence might not be converging. Coercivity implies that our minimizing sequence $\{u_k\}$ is bounded. But in infinite, bounded sequence doesn't give a convergent subsequence. So what do we do? that's when we rely on weak topology.

Since, we have assumed $q > 1$, the Sobolev Space is reflexive. and in a reflexive space, every bounded sequence has a weakly convergent subsequence,

$$u_{k_j} \rightharpoonup u(\text{lets say}).$$

Then again, there is an issue. $I[\cdot]$ is not continuous w.r.t weak convergence. To see this, take $u_k(x) = \sin(kx)$

To show that $u_k(x) \rightharpoonup 0$ in $L^2([0, 1])$, we have to show that

$$\lim_{k \rightarrow \infty} \int_0^1 \phi(x) \sin(kx) dx = 0$$

$$\int_0^1 \sin(kx) \phi(x) \stackrel{\text{int-by-parts}}{=} \int_0^1 -\frac{1}{k} \cos(kx) \phi'(x) \rightarrow 0 \text{ as } k \rightarrow \infty$$

So $u_k \rightharpoonup 0$ in $L^2([0, 2])$. Let $f(t) = t^2$. $f(u_k) = \sin^2(kx) = \frac{1 - \cos(2kx)}{2}$. Therefore $f(u_k(x)) \rightharpoonup \frac{1}{2} \neq 0$.

Hence, weak convergence is not enough to pass through a nonlinear function, even when that function is continuous.

So, we can't deduce

$$I[u] = \lim_{j \rightarrow \infty} I[u_{k_j}] \tag{1.16}$$

But we don't need convergence. It would suffice to know

$$I[u] \leq \liminf_{j \rightarrow \infty} I[u_{k_j}] \quad (1.17)$$

Definition 1.3.1. we say that a function $I[\cdot]$ is sequentially weakly lower semicontinuous on $W^{1,q}(U)$, provided

$$I[u] \leq \liminf_{j \rightarrow \infty} I[u_k]$$

whenever

$$u_k \rightharpoonup u \quad \text{weakly in } W^{1,q}(U)$$

From second variation section, we have

$$\sum_{i,j=1}^n L_{p_i p_j}(Du(x), u(x), x) \xi_i \xi_j \geq 0, \quad \xi \in \mathbb{R}^n, x \in U$$

This inequality suggests that it is reasonable to assume that L is convex in its first argument.

Theorem 1.3.2 (Weak lower semicontinuity).

Assume that L is smooth, bounded below and in addition

the mapping $p \mapsto L(p, z, x)$ is convex

for each $z \in \mathbb{R}, x \in U$. Then

$I[\cdot]$ is weakly lower semicontinuous on $W^{1,q}(U)$.

Proof. Let $\{u_k\}_{k=1}^\infty$ be a sequence with

$$u_k \rightharpoonup u \quad \text{weakly in } W^{1,q}(U), \quad (1.18)$$

Claim:

$$I[u] \leq m = \liminf_{k \rightarrow \infty} I[u_k] \quad (1.19)$$

Note that since any weakly convergent sequence is bounded, we have

$$\sup_k \|u_k\|_{1,q,U} < \infty \quad (1.20)$$

Since $l := \liminf_{k \rightarrow \infty} I[u_k]$, by the definition of $\liminf$ there exists a subsequence $\{u_{k_j}\}_{j \in \mathbb{N}}$ such that $I[u_{k_j}] \rightarrow l$ as $j \rightarrow \infty$. To simplify notation, we relabel this subsequence as $\{u_k\}_{k \in \mathbb{N}}$. Hence, after passing to this subsequence and relabelling, we may assume without loss of generality that

$$l = \lim_{k \rightarrow \infty} I[u_k].$$

From the compactness theorem,

$$u_k \rightarrow u \quad \text{strongly in } L^q(U) \quad (1.21)$$

Theorem 4.9 from Brezis's says

Theorem 4.9. Let (f_n) be a sequence in L^p and let $f \in L^p$ be such that $\|f_n - f\|_p \rightarrow 0$.

Then, there exist a subsequence (f_{n_k}) and a function $h \in L^p$ such that

- (a) $f_{n_k}(x) \rightarrow f(x)$ a.e. on Ω ,
- (b) $|f_{n_k}(x)| \leq h(x), \quad \forall k, \text{ a.e. on } \Omega.$

So, passing if necessary to yet another subsequence,

$$u_k \rightarrow u \quad \text{a.e. in } U. \quad (1.22)$$

Fix $\epsilon > 0$. We recall Egoroff's Theorem

THEOREM 2 (Egoroff's Theorem). Let $\{f_k\}_{k=1}^\infty, f$ be measurable functions, and assume

$$f_k \rightarrow f \quad \text{a.e. on } A,$$

where $A \subset \mathbb{R}^n$ is measurable, $|A| < \infty$. Then for each $\varepsilon > 0$ there exists a measurable subset $E \subset A$ such that

$$(i) \quad |A - E| \leq \varepsilon$$

and

$$(ii) \quad f_k \rightarrow f \quad \text{uniformly on } E.$$

1.22 and Egoroff's Theorem imply

$$u_k \rightarrow u \quad \text{uniformly on } E_\epsilon \tag{1.23}$$

where E_ϵ is a measurable set with

$$|U - E_\epsilon| \leq \epsilon \tag{1.24}$$

Assume $E_\epsilon \subset E_{\epsilon'}$ for $0 < \epsilon' < \epsilon$. Set

$$F_\epsilon = \{x \in U \mid |u(x) + |Du(x)|| \leq 1/\epsilon\} \tag{1.25}$$

So,

$$|U - F_\epsilon| = |\{x \in U \mid |u(x) + |Du(x)|| > 1/\epsilon\}| \stackrel{\text{Markov's inequality}}{\leq} \epsilon \| |u(x) + |Du(x)|| \|_{L^1} \tag{1.26}$$

where, Markov's inequality states

4.1 Markov's inequality

Suppose $(X, \mathcal{S}, \mu)$ is a measure space and $h \in L^1(\mu)$. Then

$$\mu(\{x \in X : |h(x)| \geq c\}) \leq \frac{1}{c} \|h\|_1$$

for every $c > 0$.

set

$$G_\epsilon = E_\epsilon \cap F_\epsilon \tag{1.27}$$

from 1.24 and 1.26, $|U - G_\epsilon| \rightarrow 0$ as $\epsilon \rightarrow 0$.

Since, L is bounded, we may as well assume that

$$L \geq 0 \tag{1.28}$$

Consequently,

$$\begin{aligned} I[u_k] &= \int_U L(Du_k, u_k, x) \, dx \geq \int_{G_\epsilon} L(Du_k, u_k, x) \\ &\stackrel{L \text{ is convex in } p}{\geq} \int_{G_\epsilon} L(Du, u_k, x) \, dx + \int_{G_\epsilon} D_p L(Du, u_k, x) \cdot (Du_k - Du) \, dx \end{aligned} \tag{1.29}$$

1.23, 1.25 and 1.27 imply

$$\lim_{k \rightarrow \infty} \int_{G_\epsilon} L(Du, u_k, x) \, dx = \int_{G_\epsilon} L(Du, u, x) \, dx \tag{1.30}$$

Since $D_p L(Du, u_k, x) \rightarrow D_p L(Du, u, x)$ uniformly on G_ϵ and $Du_k \rightarrow Du$ weakly in $L^q(U; \mathbb{R}^n)$,

$$\lim_{k \rightarrow \infty} \int_{G_\epsilon} D_p L(Du, u_k, x) \cdot (Du_k - Du) dx = 0. \quad (1.31)$$

1.29, 1.30 and 1.31 imply

$$m = \lim_{k \rightarrow \infty} I[u_k] \geq \int_{G_\epsilon} L(Du, u, x) dx = \int_U L(Du, u, x) \mathbf{1}_{G_\epsilon}(x) dx.$$

where,

$$\mathbf{1}_{G_\epsilon}(x) = \begin{cases} 1, & x \in G_\epsilon, \\ 0, & x \notin G_\epsilon. \end{cases}$$

As $\epsilon \rightarrow 0$, $G_\epsilon \uparrow U$,

and hence, for $\epsilon_1 > \epsilon_2 > 0$, $L(Du, u, x) \mathbf{1}_{G_{\epsilon_1}}(x) \leq L(Du, u, x) \mathbf{1}_{G_{\epsilon_2}}(x)$ for a.e. $x \in U$.

Therefore, by the Monotone Convergence Theorem,

$$\lim_{\epsilon \rightarrow 0} \int_U L(Du, u, x) \mathbf{1}_{G_\epsilon}(x) dx = \int_U L(Du, u, x) dx.$$

□

Theorem 1.3.3 (Existence of minimizer).

Assume that L satisfies the coercivity inequality 1.14, and is convex in the variable p . Suppose also that the admissible set $\mathcal{A}$.

Then there exists at least one function $u \in \mathcal{A}$ solving

$$I[u] = \min_{w \in \mathcal{A}} I[w]$$

Proof. Let $\{u_k\}_{k=1}^\infty \subset \mathcal{A}$ be a minimizing sequence satisfying

$$I[u_k] \rightarrow m = \inf_{w \in \mathcal{A}} I[w] \quad (1.32)$$

Take $\beta = 0$ in inequality 1.14, otherwise consider $\tilde{L} = L + \beta$. Thus, $L \geq \alpha|p|^q$, and so

$$I[w] \geq \alpha \int_U |Dw|^q dx \quad (1.33)$$

If m is finite, then 1.32 and 1.33 imply that

$$\sup_k \|Du_k\|_{q,U} < \infty \quad (1.34)$$

Now, we fix a function $w \in \mathcal{A}$. Since both u_k and w equal to g on ∂U in the trace sense, we get $u_k - w \in W_0^{1,q}(U)$.

We recall Poincaré's inequality:

$$\int_U |u|^2 \leq C \int_U |Du|^2 dx, \quad \forall u \in W_0^{1,q}(U), \quad 1 \leq q, \infty \quad (1.35)$$

We get

$$\begin{aligned} \|u_k\|_{q,U} &\leq \|u_k - w\|_{q,U} + \|w\|_{q,U} \\ &\leq C \|Du_k - Dw\|_{q,U} + \|Dw\| \stackrel{1.34}{\leq} C \end{aligned}$$

Hence, $\sup_k \|u_k\|_{q,U} < \infty$.

This estimate and 1.34 imply that $\{u_k\}_{k=1}^\infty$ is bounded in $W^{1,q}(U)$.

Consequently, we can find a subsequence $\{u_{k_j}\}_{j=1}^\infty \subset \{u_k\}_{k=1}^\infty$ and a function $u \in W^{1,q}(U)$ such that

$$u_{k_j} \rightarrow u \quad \text{in } W^{1,q}(U).$$

Claim: $u \in \mathcal{A}$

To prove this, we first recall Mazur's theorem:

A convex, closed subspace of a Banach space X is weakly closed.

Now, note that for $w \in \mathcal{A}$, $u_k - w \in W_0^{1,q}(U)$. $W_0^{1,q}(U)$ is a closed, linear subspace of $W^{1,q}(U)$. By Mazur's theorem, $W_0^{1,q}(U)$ is weakly closed.

Hence, $u - w \in W_0^{1,q}(U)$. Consequently, trace of u on ∂U is g .

Since, L is coercive, and convex in variable p , Theorem 1.3.2 imply that I is lower semi continuous, and $I[u] \leq \liminf_{j \rightarrow \infty} I[u_{k_j}] = m$. But since $u \in \mathcal{A}$, we have

$$I[u] = m = \min_{w \in \mathcal{A}} I[w].$$

□

In order to prove uniqueness, we need further assumptions. Suppose

$$L = L(p, x) \text{ does not depend on } z \quad (1.36)$$

and

$$\begin{cases} \exists \theta > 0 \text{ such that} \\ \sum_{i,j=1}^n L_{p_i p_j}(p, x) \xi_i \xi_j \geq \theta |\xi|^2, \quad p, \xi \in \mathbb{R}^n, x \in U \end{cases} \quad (1.37)$$

Equation 1.37 implies that the mapping $p \mapsto L(p, x)$ is uniformly convex for each x .

Theorem 1.3.4 (Uniqueness of minimizer).

Suppose 1.36, 1.37 hold. Then a minimizer $u \in \mathcal{A}$ of $I[\cdot]$ is unique.

Proof. Let u and $\tilde{u}$ both be minimizers of $I[\cdot]$ over $\mathcal{A}$. Then $v = \frac{u+\tilde{u}}{2} \in \mathcal{A}$.

Claim:

$$I[v] \leq \frac{I[u] + I[\tilde{u}]}{2} \quad (1.38)$$

From Fundamental Theorem of Calculus,

$$\begin{aligned} L(p, x) - L(q, x) &= \int_0^1 DL(q + t(p - q), x) \cdot (p - q) dt \\ \implies L(p, x) - L(q, x) &= \int_0^1 \left[DL(q, x) + (DL(q + t(p - q), x) - DL(q, x)) \right] \cdot (p - q) \\ \implies L(p, x) - L(q, x) &= DL(q, x) \cdot (p - q) + \int_0^1 \left[DL(q + t(p - q), x) - DL(q, x) \right] \cdot (p - q) \\ \implies L(p, x) - L(q, x) &= DL(q, x) \cdot (p - q) + \int_0^1 \int_0^t D^2 L(q + s(p - q), x) (p - q) \cdot (p - q) ds dt \\ \implies L(p, x) - L(q, x) &\geq DL(q, x) \cdot (p - q) + \int_0^1 \int_0^t \theta |p - q|^2 ds dt \\ \implies L(p, x) &\geq L(q, x) + DL(q, x) \cdot (p - q) + \frac{\theta}{2} |p - q|^2, \quad x \in U, p, q \in \mathbb{R}^n \end{aligned} \quad (1.39)$$

Set $q = \frac{Du + D\tilde{u}}{2}$, $p = Du$ and integrating over U ,

$$\begin{aligned} I[u] &\geq I[v] + \int_U DL\left(\frac{Du + D\tilde{u}}{2}, x\right) \cdot \left(\frac{Du - D\tilde{u}}{2}\right) dx \\ &\quad + \frac{\theta}{8} \int_U |Du - D\tilde{u}|^2 dx \end{aligned} \quad (1.40)$$

Similarly, for $q = \frac{Du + D\tilde{u}}{2}$, $p = D\tilde{u}$ in equation 1.39 and integrating over U ,

$$\begin{aligned} I[\tilde{u}] \geq I[v] + \int_U DL\left(\frac{Du + D\tilde{u}}{2}, x\right) \cdot \left(\frac{D\tilde{u} - Du}{2}\right) dx \\ + \frac{\theta}{8} \int_U |Du - D\tilde{u}| dx \end{aligned} \quad (1.41)$$

Adding equations 1.40 and 1.41 and dividing by 2,

$$I[v] + \frac{\theta}{8} \int_U |Du - D\tilde{u}| dx \leq \frac{I[u] + I[\tilde{u}]}{2}$$

Since, u and $\tilde{u}$ are both minimizers,

$$I[u] = I[\tilde{u}] = \min_{w \in \mathcal{A}} I[w] \leq I[v], \quad \text{since } v \in \mathcal{A}.$$

So,

$$\begin{aligned} I[u] &\leq I[v] + \frac{\theta}{8} \int_U |Du - D\tilde{u}| dx \leq \frac{I[u] + I[\tilde{u}]}{2} = I[u] \\ &\implies \int_U |Du - D\tilde{u}| dx = 0 \\ &\implies Du = D\tilde{u} \text{ a.e. in } U \end{aligned}$$

Now, define $w = u - \tilde{u} \in H_0^1(U)$. Then $Dw = Du - D\tilde{u} = 0$ a.e. in U . By Poincaré inequality, $w = 0$ a.e. This implies $u = \tilde{u}$ a.e. in U . $\square$

1.3.1 Weak solutions of Euler-Lagrange equation.

In this section, we

Suppose

$$|L(p, z, x)| \leq C(|p|^q + |z|^q + 1) \quad (1.42)$$

and

$$\begin{cases} |D_p L(p, z, x)| \leq C(|p|^{q-1} + |z|^{q-1} + 1) \\ |D_z L(p, z, x)| \leq C(|p|^{q-1} + |z|^{q-1} + 1) \end{cases} \quad (1.43)$$

for some constant C and all $p \in \mathbb{R}^n$, $z \in \mathbb{R}$, $x \in U$

Motivation for definition of weak solution.

We consider a boundary value problem for the Euler-Lagrange equation associated with functional L , and which for a smooth minimizer u reads

$$\begin{cases} -\sum_{i=1}^n (L_{p_i}(Du, u, x))_{x_i} + L_z(Du, u, x) = 0, & \text{in } U \\ u = g, & \text{on } \partial U \end{cases} \quad (1.44)$$

Multiplying 1.44 by $v \in C_0^\infty(U)$ and applying integrating by parts,

$$\int_U \sum_{i=1}^n L_{p_i}(Du, u, x) v_{x_i} + L_z(Du, u, x) v \, dx = 0 \quad (1.45)$$

Note, that 1.45 is valid for smooth functions u . But if assume $u \in W^{1,q}(U)$, then we come across questions regarding the integrability of L_{p_i} and L_z . And that's where our growth assumptions, 1.42 and 1.43, come into play.

We see that

$$|D_p L(Du, u, x)| \leq C(|Du|^{q-1} + |u|^{q-1} + 1) \in L^{q'}(U), \quad (1.46)$$

where q' is the Hölder conjugate of q . Similarly,

$$|D_z L(Du, u, x)| \leq C(|Du|^{q-1} + |u|^{q-1} + 1) \in L^{q'}(U), \quad (1.47)$$

By density theorem, we see that 1.45 is valid for any $v \in W_0^{1,q}(U)$.

Definition 1.3.5. We say $u \in \mathcal{A}$ is a weak solution of the boundary-value problem 1.44 for the Euler-Lagrange equation provided

$$\int_U \sum_{i=1}^n L_{p_i}(Du, u, x) v_{x_i} + L_z(Du, u, x) v \, dx = 0, \quad \forall v \in W_0^{1,q}(U).$$

Theorem 1.3.6 (Solutions of Euler-Lagrange equation).

Assume that L satisfies the growth conditions 1.42, 1.43 and $u \in \mathcal{A}$ satisfies

$$I[u] = \min_{w \in \mathcal{A}} I[w]$$

Then u is a weak solution of 1.44

Proof. Fix any $v \in W_0^{1,q}(U)$, and define

$$i(\tau) = I[u + \tau v], \quad \tau \in \mathbb{R}$$

By 1.42, $i(\cdot)$ is finite for all τ .

Let $\tau \neq 0$, then

$$\frac{i(\tau) - i(0)}{\tau} = \int_U \frac{L(Du + \tau Dv, u + \tau v, x) - L(Du, u, x)}{\tau} dx = \int_U L^\tau(x) dx, \quad (1.48)$$

where

$$L^\tau(x) = \frac{1}{\tau} [L(Du(x) + \tau Dv(x), u(x) + \tau v(x), x) - L(Du(x), u(x), x)], \quad a.e. x \in U$$

Observe that

$$L^\tau(x) \rightarrow \sum_{i=1}^n L_{p_i}(Du, u, x) v_{x_i} + L_z(Du, u, x) v \text{ as } \tau \rightarrow 0 \quad (1.49)$$

and

$$\begin{aligned} L^\tau(x) &\stackrel{F.T.C}{=} \frac{1}{\tau} \int_0^\tau \frac{d}{ds} L(Du + sDv, u + sv, x) \\ &= \frac{1}{\tau} \int_0^\tau \sum_{i=1}^n L_{p_i}(Du + sDv, u + sv, x) v_{x_i} + L_z(Du + sDv, u + sv, x) v ds \\ &\implies |L^\tau(x)| \leq \frac{1}{|\tau|} \int_0^{|\tau|} |D_p L(Du + sDv, u + sv, x)| |Dv| \\ &\quad + |D_z L(Du + sDv, u + sv, x)| |v| ds \\ &\stackrel{1.43}{\leq} \frac{1}{|\tau|} \int_0^{|\tau|} C (|Du|^{q-1} + (|s| |Dv|)^{q-1} + |u|^{q-1} + (|s| |v|)^{q-1} + 1) (|Dv| + |v|) \\ &\leq C (|Du|^{q-1} + |Dv|^{q-1} + |u|^{q-1} + |v|^{q-1} + 1) (|Dv| + |v|) \\ &\stackrel{\text{Young's ineq}}{\leq} C (|Du|^q + |u|^q + |Dv|^q + |v|^q + 1) \in L^1(U) \end{aligned}$$

By Dominated Convergence Theorem,

$$\begin{aligned} \lim_{\tau \rightarrow 0} \int_U L^\tau(x) dx &= \int_U \sum_{i=1}^n L_{p_i}(Du, u, x)v_{x_i} + L_z(Du, u, x)v dx \\ \implies \text{from 1.48, } i'(0) &= \int_U \sum_{i=1}^n L_{p_i}(Du, u, x)v_{x_i} + L_z(Du, u, x)v dx \end{aligned}$$

Since $i(\cdot)$ has a minimum at $\tau = 0$, $i'(0) = 0$. Thus, u is a weak solution. □

Remark 1.3.7. Generally, the Euler-Lagrange equation 1.44 will have other solutions which are not minimizers of $I[\cdot]$.

However, if the mapping $(p, z) \mapsto L(p, z, x)$ is convex for each x , then each weak solution is a minimizer.

To see this, suppose $u \in \mathcal{A}$ solves

$$\begin{cases} -\sum_{i=1}^n (L_{p_i}(Du, u, x))_{x_i} + L_z(Du, u, x) = 0, & \text{in } U \\ u = g, & \text{on } \partial U \end{cases} \quad (1.50)$$

in the weak sense. Let $w \in \mathcal{A}$ be arbitrary.

Convexity of the mapping $(p, z) \mapsto L(p, z, x)$ gives

$$L(p, z, x) + D_p L(p, z, x) \cdot (q - p) + D_z L(p, z, x) \cdot (w - z) \leq L(q, w, x)$$

Letting $p = Du(x)$, $q = Dw(x)$, $z = u(x)$, $w = w(x)$ and integrating over U gives

$$I[u] + \int_U D_p L(Du, u, x) \cdot (Dw - Du) + D_z L(Du, u, x)(w - u) dx \leq I[w]$$

Observe that $w - u \in W_0^{1,q}(U)$. And since u is a weak solution of the Euler-Lagrange equation, the second term on the left is zero. Therefore,

$$I[u] \leq I[w], \quad \text{for each } w \in \mathcal{A}.$$

1.4 Regularity

We suppose that the functional $I[\cdot]$ has the form

$$I[w] = \int_U L(Dw) - wf \, dx, \quad f \in L^2(U) \quad (1.51)$$

We take $q = 2$, and suppose the growth condition

$$|D_p L(p)| \leq C(|p| + 1), \quad p \in \mathbb{R}^n \quad (1.52)$$

Then any minimizer $u \in \mathcal{A}$ is a weak solution of the Euler-Lagrange PDE

$$-\sum_{i=1}^n (L_{p_i}(Du))_{x_i} = f, \quad \text{in } U; \quad (1.53)$$

or, in weak form

$$\int_U \sum_{i=1}^n L_{p_i}(Du) v_{x_i} \, dx = \int_U f v \, dx, \quad \forall v \in H_0^1(U). \quad (1.54)$$

We now want to show that if $u \in H^1(U)$ is a weak solution of 1.53, then $u \in H_{loc}^2(U)$. To establish this, we need a stronger growth condition on L . Suppose

$$|D^2 L(p)| \leq C, \quad p \in \mathbb{R}^n \quad (1.55)$$

We also assume that L is uniformly convex,

$$\sum_{i,j=1}^n L_{p_i p_j}(p) \xi_i \xi_j \geq \theta |\xi|^2, \quad p, \xi \in \mathbb{R}^n; \text{ constant } \theta > 0 \quad (1.56)$$

Theorem 1.4.1 (Second derivatives for minimizers). *(i) Let $u \in H^1(U)$ be a weak solution of the nonlinear partial differential equation 1.53, where L satisfies 1.55 and 1.56. Then*

$$u \in H_{loc}^2(U)$$

(ii) If $u \in H_0^1(U)$ and ∂U is C^2 , then

$$u \in H^2(U)$$

with estimate

$$\|u\|_{H^2(U)} \leq C \|f\|_{L^2(U)}$$

Proof. Choose open sets V and U such that $V \subset\subset W \subset\subset U$. Let ζ be a smooth cut off function satisfying

$$\zeta \equiv 1 \text{ on } V, \zeta \equiv 0 \text{ in } \mathbb{R}^n - W, 0 \leq \zeta \leq 1$$

Let $|h| > 0$ be small and substitute

$$v = -D_k^{-h}(\zeta^2 D_k^h u), \quad k \in \{1, 2, \dots, n\}$$

into 1.54. We recall the notation for difference quotient

$$D_k^h u(x) = \frac{u(x + he_k) - u(x)}{h}, \quad x \in W$$

We also use the identity

$$\int_U u D_k^{-h} v \, dx = - \int_U v D_k^h u \, dx \quad (1.57)$$

we deduce

$$\begin{aligned} & \int_U \sum_{i=1}^n L_{p_i}(Du)(-D_k^{-h}(\zeta^2 D_k^h u))_{x_i} \, dx = \int_U f(-D_k^{-h}(\zeta^2 D_k^h u)) \, dx \\ &= \sum_{i=1}^n \int_U L_{p_i}(Du)(-D_k^{-h}(\zeta^2 D_k^h u))_{x_i} \, dx = \int_U f(-D_k^{-h}(\zeta^2 D_k^h u)) \, dx \quad (1.58) \\ &\stackrel{1.57}{=} \sum_{i=1}^n \int_U D_k^h(L_{p_i}(Du))(\zeta^2 D_k^h u)_{x_i} \, dx = \int_U f(-D_k^{-h}(\zeta^2 D_k^h u)) \, dx \end{aligned}$$

Now,

$$\begin{aligned} D_k^h(L_{p_i}(Du(x))) &= \frac{L_{p_i}(Du(x + he_k)) - L_{p_i}(Du(x))}{h} \\ &\stackrel{\text{F.T.C}}{=} \frac{1}{h} \int_0^1 \frac{d}{ds} L_{p_i}(sDu(x + he_k) + (1-s)Du(x)) \, ds \\ &\stackrel{\text{chain rule}}{=} \frac{1}{h} \int_0^1 \sum_{j=1}^n L_{p_i p_j}(sDu(x + he_k) + (1-s)Du(x)) (u_{x_j}(x + he_k) - u_{x_j}(x)) \, ds \\ &= \sum_{j=1}^n a^{ij,h}(x) D_k^h u_{x_j}(x) \quad (1.59) \end{aligned}$$

for

$$a^{ij,h}(x) = \int_0^1 L_{p_i p_j}(sDu(x + he_k) + (1-s)Du(x)) ds, \quad i, j \in \{1, 2, \dots, n\} \quad (1.60)$$

We substitute 1.59 into 1.58:

$$\begin{aligned} A_1 + A_2 &= \sum_{i,j=1}^n \int_U a^{ij,h} D_k^h u_{x_j} \zeta^2 D_k^h u_{x_i} dx \\ &\quad + \sum_{i,j=1}^n \int_U a^{ij,h} D_k^h u_{x_j} D_k^h u \ 2\zeta \zeta_{x_i} dx \\ &= - \int_U f(-D_k^{-h}(\zeta^2 D_k^h u)) dx = B \end{aligned} \quad (1.61)$$

Applying uniform convexity 1.56 to A_1 ,

$$A_1 \geq \theta \int_U \zeta^2 |D_k^h Du|^2 dx \quad (1.62)$$

For A_2 :

$$\begin{aligned} |A_2| &\leq \sum_{i,j=1}^n \int_U \left(\int_0^1 |L_{p_i p_j}(\dots)| ds \right) |D_k^h u_{x_j}| |D_k^h u| \ 2\zeta |\zeta_{x_i}| dx \\ &\stackrel{1.55}{\leq} C \int_U |D_k^h Du| |D_k^h u| \zeta dx \\ &\stackrel{\zeta \equiv 0 \text{ in } \mathbb{R}^n - W}{\leq} C \int_W |D_k^h Du| |D_k^h u| \zeta dx \\ &\stackrel{\text{Cauchy's inequality with } \epsilon}{\leq} \epsilon \int_W \zeta^2 |D_k^h Du|^2 dx + \frac{C}{\epsilon} \int_W |D_k^h u|^2 dx \end{aligned} \quad (1.63)$$

For B ,

$$\begin{aligned} |B| &\leq \int_U |f| |D_k^{-h}(\zeta^2 D_k^h u)| dx \\ &\stackrel{\text{Cauchy's inequality with } \epsilon}{\leq} \epsilon \int_U |D_k^{-h}(\zeta^2 D_k^h u)|^2 dx + \frac{C}{\epsilon} \int_U |f|^2 dx \end{aligned} \quad (1.64)$$

Now,

$$\begin{aligned}
\int_U |D_k^{-h}(\zeta^2 D_k^h u)|^2 dx &= \int_{W_h} |D_k^{-h}(\zeta^2 D_k^h u)|^2 dx, \quad W_h = \{x \in U \mid \text{dist}(x, W) < |h|\} \subset\subset U \\
&\stackrel{\text{theorem 3(i) 5.8.1}}{\leq} C \int_U |D(\zeta^2 D_k^h u)|^2 dx \\
&= C \int_U |2\zeta D\zeta D_k^h u + \zeta^2 D_k^h Du|^2 dx \\
&\leq C \int_U \zeta^2 |D\zeta|^2 |D_k^h u|^2 dx + C \int_U \zeta^4 |D_k^h Du|^2 dx \\
&\leq C \int_W |D_k^h u|^2 dx + C \int_U \zeta^2 |D_k^h Du|^2 dx \\
&\stackrel{\text{theorem 3(i) 5.8.1}}{\leq} C \int_U |Du|^2 dx + C \int_U \zeta^2 |D_k^h Du|^2 dx.
\end{aligned}$$

1.64 becomes

$$|B| \leq \epsilon \int_U \zeta^2 |D_k^h Du|^2 dx + \frac{C}{\epsilon} \int_U f^2 + |Du|^2 dx$$

Combining all the bounds on A_1 , A_2 and B ,

$$\begin{aligned}
|A_1| &\leq |A_2| + |B| \\
\implies \theta \int_U \zeta^2 |D_k^h Du|^2 dx &\leq \left(\epsilon \int_U \zeta^2 |D_k^h Du|^2 dx + \frac{C}{\epsilon} \int_W |D_k^h u|^2 dx \right) \\
&\quad + \left(\epsilon \int_U \zeta^2 |D_k^h Du|^2 dx + \frac{C}{\epsilon} \left(\int_U f^2 + |Du|^2 dx \right) \right) \\
\implies (\theta - 2\epsilon) \int_U \zeta^2 |D_k^h Du|^2 dx &\leq \frac{C}{\epsilon} \left(\int_U f^2 + |Du|^2 dx + \int_W |D_k^h u|^2 dx \right) \\
&\stackrel{\epsilon=\theta/4}{\implies} \frac{\theta}{2} \int_U \zeta^2 |D_k^h Du|^2 dx \leq \frac{C}{\theta} \left(\int_U f^2 + |Du|^2 dx + \int_W |D_k^h u|^2 dx \right) \\
&\stackrel{\text{Thm 3(i) on } u}{\implies} \int_U \zeta^2 |D_k^h Du|^2 dx \leq C \left(\int_U f^2 + |Du|^2 dx \right),
\end{aligned}$$

Since $\zeta \equiv 1$ on V ,

$$\int_V |D_k^h Du|^2 dx \leq C \left(\int_U f^2 + |Du|^2 dx \right), \quad k \in \{1, \dots, n\} \text{ and sufficiently small } |h| > 0.$$

Theorem 3 (ii) in 5.8.1 implies $Du \in H^1(U) \implies u \in H^2(U)$. Since $V \subset\subset U$, we

get, $u \in H_{loc}^2(U)$.

□

Proof. Now, for boundary case

Case 1: U is a half ball:

$$U = B^0(0, 1) \cap \mathbb{R}_+^n \quad (1.65)$$

Set $V = B^0(0, \frac{1}{2}) \cap \mathbb{R}_+^n$. Let ζ be a smooth cut-off function satisfying

$$\begin{cases} \zeta = 1 \text{ on } B(0, \frac{1}{2}), \zeta \equiv 0 \text{ on } \mathbb{R}^n - B(0, 1) \\ 0 \leq \zeta \leq 1 \end{cases}$$

So $\zeta \equiv 1$ on V and ζ vanishes bnear the curved part of ∂U . Let $h > 0$ be small and define

$$v = -D_k^{-1}(\zeta^2 D_k^h u), \quad k \in \{1, \dots, n-1\}$$

We note that

$$\begin{aligned} v(x) &= -\frac{1}{h} D_k^{-h}(\zeta^2(x)[u(x + he_k) - u(x)]) \\ &= \frac{1}{h^2}(\zeta^2(x - he_k)[u(x) - u(x - he_k)] - \zeta^2(x)[u(x + he_k) - u(x)]) \end{aligned}$$

Since $u = 0$ along $\{x_n = 0\}$ in the trace sense and $\zeta \equiv 0$ near the curved portion of ∂U , we see $v \in H_0^1(U)$.

Reason: Why did we took $u = 0$ along $\{x_n = 0\}$ if we had already assumed $x \in U = B^0(0, 1) \cap \mathbb{R}_+^n$?

for $k \in \{1, 2, \dots, n-1\}$, both x and $x + he_k$ lie on the flat boundary, $\{x_n = 0\}$. Therefore,

$$u(x) = u(x + he_k) = 0.$$

And

$$D_k^h u(x) = 0.$$

But on the curved part of the boudary of $B^0(0, 1) \cap \mathbb{R}_+^n$, for $k \in \{1, 2, \dots, n-1\}$, $x + he_k$ may shift tangentially to left or right. So it may not lie on the curved boundary of U . Therefore,

$$u(x + he_k) \neq 0, \quad \text{on the courved part of the boundary of } U.$$

That's why we took $u = 0$ along $\{x_n = 0\}$.

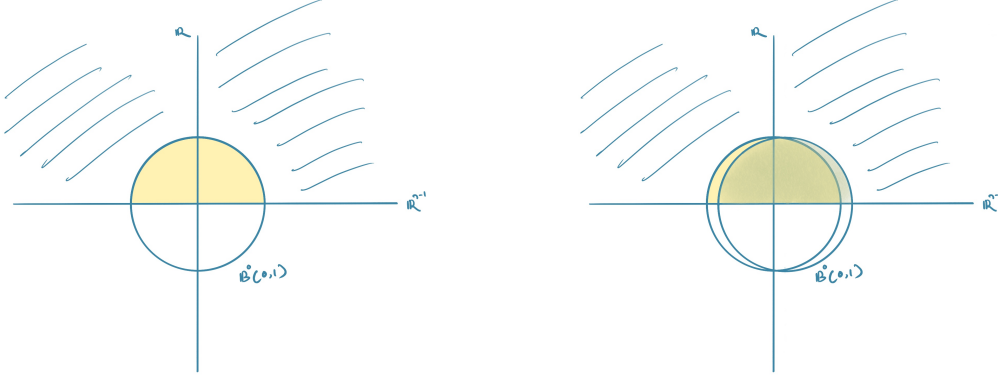


Figure 1.4.1: x and $x + he_k$ lie on the flat boundary but $x + he_k$ doesn't lie in U .

Like the previous proof, we have the estimate

$$\int_V |D_k^h Du|^2 dx \leq C \int_U f^2 + |Du|^2 dx, \quad k \in \{1, \dots, n-1\}$$

Theorem 3 (ii) in 5.8.1 implies

$$u_{x_k} \in H^1(V), \quad k \in \{1, \dots, n-1\}$$

with the estimate

$$\sum_{\substack{k,l=1 \\ k+l \leq 2n}}^n \|u_{x_k x_l}\|_{L^2(V)} \leq C \left(\|f\|_{L^2(U)} + \|u\|_{H^1(U)} \right) \quad (1.66)$$

We must augment 1.66 with an estimate of the L^2 norm of $u_{x_n x_n}$ over V .

We recall 1.53

$$\begin{aligned}
& -\sum_{i=1}^n (L_{p_i}(Du))_{x_i} = f, \quad \text{in } U \\
& -\sum_{i,j=1}^n L_{p_i p_j}(Du(x))u_{x_i x_j} = f \\
\implies & -L_{p_n p_n}(Du(x))u_{x_n x_n} = f + \sum_{\substack{i,j=1 \\ i+j < 2n}}^n L_{p_i p_j}(Du(x))u_{x_i x_j} \quad (1.67)
\end{aligned}$$

Uniform convexity implies that

$$\sum_{i,j=1}^n L_{p_i p_j}(p)\xi_i \xi_j \geq \theta |\xi|^2, \quad p, \xi \in \mathbb{R}^n; \text{ constant } \theta > 0, \quad x \in U, \xi \in \mathbb{R}^n$$

Set $\xi = e_n = (0, \dots, 0, 1)$ to conclude

$$L_{p_n p_n}(p) \geq \theta \quad (1.68)$$

Also, from equation 1.55,

$$\begin{aligned}
& |D^2 L| \leq C \\
\implies & |L_{p_i p_j}(Du)| \leq C
\end{aligned}$$

Equation 1.67 gives

$$\begin{aligned}
& u_{x_n x_n} = -\frac{1}{L_{p_n p_n}(Du(x))} \left(f + \sum_{\substack{i,j=1 \\ i+j < 2n}}^n L_{p_i p_j}(Du(x))u_{x_i x_j} \right) \\
\implies & |u_{x_n x_n}| \leq \frac{1}{|L_{p_n p_n}(Du(x))|} \left(|f| + \sum_{\substack{i,j=1 \\ i+j < 2n}}^n |L_{p_i p_j}(Du(x))u_{x_i x_j}| \right) \\
& \implies |u_{x_n x_n}| \leq C \left(|f| + \sum_{\substack{i,j=1 \\ i+j < 2n}}^n |u_{x_i x_j}| \right) \\
& \implies \|u_{x_n x_n}\|_{L^2(V)} \leq C \left(\|f\|_{L^2(V)} + \sum_{\substack{i,j=1 \\ i+j < 2n}}^n \|u_{x_i x_j}\|_{L^2(V)} \right) \quad (1.69)
\end{aligned}$$

Utilizing this estimate in inequality 1.66, we conclude $u \in H^2(V)$, and

$$\|u\|_{H^2(V)} \leq C(\|f\|_{L^2(U)} + \|u\|_{H^1(U)}) \quad (1.70)$$

for some constant C .

General Case: We choose any point $x^0 \in \partial U$ and since ∂U is C^2 , we assume that

$$U \cap B(x^0, r) = \{x \in B(x^0, r) \mid x_n > \gamma(x_1, \dots, x_{n-1})\}$$

for some $r > 0$, and C^2 function $\gamma : \mathbb{R}^{n-1} \rightarrow \mathbb{R}$. Set

$$y = \Phi(x), \quad x = \Psi(y). \quad (1.71)$$

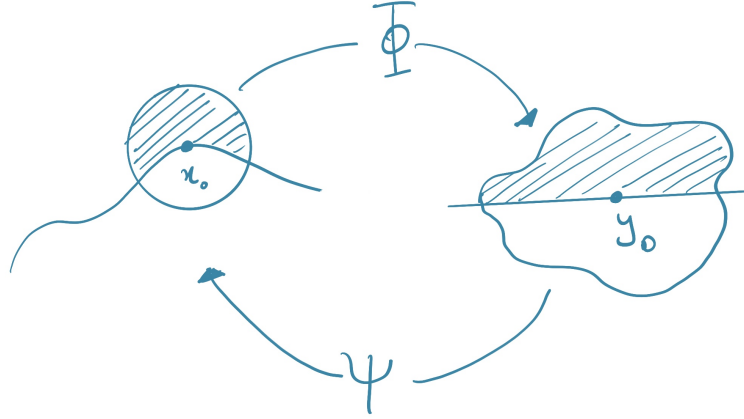


Figure 1.4.2: Straightening of the boundary.

Choose sufficiently small $s > 0$ such that $U' = B_0(0, s) \cap \{y_n > 0\}$ lies in $\Phi(U \cap B(x^0, r))$. Set $V' = B^0(0, s/2) \cap \{y_n > 0\}$.

Define

$$u'(y) = u(\Psi(y)), \quad y \in U' \quad (1.72)$$

Clearly,

$$u' \in H^1(U') \quad (1.73)$$

and

$$u' = 0 \text{ on } \partial U' \cap \{y_n = 0\} \quad (1.74)$$

in the trace sense.

Claim: u' is a weak solution of the PDE

$$-\sum_{i=1}^n (L'_{p_i}(Du'))_{y_i} = f', \quad \text{in } U \quad (1.75)$$

for

$$f'(y) = f(\Psi(y)) \quad (1.76)$$

We observe that

$$y = \Phi(x) \quad \begin{cases} y_1 = \Phi^1(x_1, \dots, x_n), \\ y_2 = \Phi^2(x_1, \dots, x_n), \\ \vdots \\ y_n = \Phi^n(x_1, \dots, x_n) \end{cases}$$

So,

$$\begin{aligned} u_{x_r}(x) &= \frac{\partial}{\partial x_r} u'(\Phi(x)) = \sum_{k=1}^n \frac{\partial u'}{\partial y_k} \frac{\partial y_k}{\partial x_r} = \sum_{k=1}^n u'_{y_k} \Phi_{x_r}^k(x) \\ \implies Du(x) &= (D\Phi(x))^T Du'(\Phi(x)) \end{aligned}$$

or

$$Du(\Psi(y)) = (D\Phi(\Psi(y)))^T Du'(y)$$

Now,

$$L'(Du') = L((D\Phi(\Psi(y)))^T Du')$$

or

$$L'(p, y) = L((D\Phi(\Psi(y)))^T p)$$

By chain rule,

$$\begin{aligned} L'_{p_i}(p, y) &= \sum_{r=1}^n L_{p_r} \left(\sum_{k=1}^n p_k \Phi_{x_r}^k \right) \left(\frac{\partial}{\partial p_i} \sum_{k=1}^n p_k \Phi_{x_r}^k \right) \\ &= \sum_{r=1}^n L_{p_r} \left(\sum_{k=1}^n p_k \Phi_{x_r}^k \right) \Phi_{x_r}^i \end{aligned}$$

Hence

$$L'_{p_i}(p, y) = \sum_{r=1}^n L_{p_r}(Du(\Psi(y))) \Phi_{x_r}^i$$

Take any test function $v' \in H_0^1(U')$. Define $v(x) = v'(\Phi(x))$.

$$v_{x_r} = \sum_{i=1}^n v'_{y_i} \Phi_{x_r}^i$$

Therefore, $Dv = (D\Phi)^T Dv'$.

The original weak form is

$$\begin{aligned} \int_U L_p(Du) \cdot Dv \, dx &= \int_U f v \, dx \\ \implies \int_{U'} L_p(Du(\Psi(y))) \cdot Dv(\Psi(y)) \, dy &= \int_{U'} f(\Psi(y)) v'(y) \, dy \\ \implies \int_{U'} \sum_{r,i} L_{p_r}(Du(\Psi(y))) \Phi_{x_r}^i v'_{y_i} \, dy &= \int_{U'} f' v' \, dy \\ \implies \int_{U'} \sum_i \left(\sum_r L_{p_r}(Du(\Psi(y))) \Phi_{x_r}^i \right) v'_{y_i} \, dy &= \int_{U'} f' v' \, dy. \end{aligned}$$

Thus u' is the weak solution of the new transformed PDE. $L'(Du', u', y) = L((D\Phi(\Psi(y)))^T Du')$.

For this calculation, let's temporarily write $p = Du'$ and $q = (D\Phi(\Psi(y)))^T p$. We already found $L'_{p_i} = \sum_{r=1}^n L_{p_r}(q) \Phi_{x_r}^i$.

differentiating one more time with respect to p_j .

$$\begin{aligned}
L'_{p_i p_j} &= \frac{\partial}{\partial p_j} \left(\sum_r L_{p_r}(q) \Phi_{x_r}^i \right) \\
&= \sum_r \frac{\partial L_{p_r}(q)}{\partial p_j} \Phi_{x_r}^i \\
&= \sum_r \left(\sum_s L_{p_r p_s}(q) \frac{\partial q_s}{\partial p_j} \right) \Phi_{x_r}^i \\
&= \sum_r \left(\sum_s L_{p_r p_s}(q) \Phi_{x_s}^j \right) \Phi_{x_r}^i \quad \left(\frac{\partial q_s}{\partial p_j} = \Phi_{x_s}^j \right) \\
&= \sum_{r,s} L_{p_r p_s}(q) \Phi_{x_r}^i \Phi_{x_s}^j.
\end{aligned}$$

Claim: $\sum_{i,j} L'_{p_i p_j} \xi_i \xi_j \geq \theta' |\xi|^2$

Substituting

$$\begin{aligned}
\sum_{i,j} L'_{p_i p_j} \xi_i \xi_j &= \sum_{i,j} \sum_{r,s} L_{p_r p_s}(q) \Phi_{x_r}^i \Phi_{x_s}^j \xi_i \xi_j \\
&= \sum_{r,s} L_{p_r p_s}(q) \left(\sum_i \Phi_{x_r}^i \xi_i \right) \left(\sum_j \Phi_{x_s}^j \xi_j \right)
\end{aligned}$$

Now define $\eta_r = \sum_i \Phi_{x_r}^i \xi_i$

Note that this is exactly $\eta = D\Phi \xi$

Therefore $\sum_{i,j} L'_{p_i p_j} \xi_i \xi_j = \sum_{r,s} L_{p_r p_s}(q) \eta_r \eta_s \geq \theta |\eta|^2$

Because $D\Phi$ and $D\Psi$ are invertible and bounded,

$\xi = D\Psi D\Phi \xi$, we have $|\xi| \leq \|D\Psi\| |D\Phi \xi|$ Therefore, $|D\Phi \xi| \geq \frac{1}{\|D\Psi\|} |\xi|$

Let $c = \frac{1}{\|D\Psi\|} > 0$. Then $|D\Phi \xi| \geq c |\xi|$

Consequently, $\sum_{i,j} L'_{p_i p_j} \xi_i \xi_j \geq \theta c^2 |\xi|^2 = \theta' |\xi|^2$

So L' is uniformly convex.

Claim: $D_p^2 L'$ is bounded

$$\begin{aligned}
L'_{p_i p_j} &= \sum_{r,s} L_{p_r p_s}(q) \Phi_{x_r}^i \Phi_{x_s}^j \\
\implies |L'_{p_i p_j}| &\leq \sum_{r,s} |L_{p_r p_s}(q)| |\Phi_{x_r}^i| |\Phi_{x_s}^j| \leq C'
\end{aligned}$$

So L' satisfies the same assumptions as L :

$$|D_p^2 L'| \leq C', \quad D_p^2 L' \geq \theta' I$$

Therefore, by the argument from the previous steps, $u' \in H^2(V')$. with the bound

$$\|u'\|_{H^2(V')} \leq C(\|f'\|_{L^2(U')} + \|u'\|_{H^1(U')})$$

consequently, for $V = \Psi(V')$

$$\|u\|_{H^2(V)} \leq C(\|f\|_{L^2(U)} + \|u\|_{H^1(U)})$$

From equation 1.56, we have

$$(DL(p) - DL(0)) \cdot p \geq \theta |p|^2; \quad p \in \mathbb{R}^n$$

Putting $v = u$ in equation 1.54

$$\begin{aligned} & \int_U DL(Du) \cdot Du \, dx = \int_U f u \, dx \\ \implies & \int_U (DL(Du) - DL(0)) \cdot Du \, dx = \int_U f u \, dx \\ \implies & \theta \int_U |Du|^2 \, dx + \int_U DL(0) \cdot Du \, dx \leq \int_U f u \, dx \stackrel{C.S.-ineq}{\leq} \|f\|_{L^2(U)} \|u\|_{L^2(U)} \\ & \stackrel{\text{Poincaré}}{\leq} C \|f\|_{L^2(U)} \|Du\|_{L^2(U)} \end{aligned}$$

We observe that

$$\begin{aligned} \int_U DL(0) \cdot Du \, dx &= \sum_{i=1}^n L_{p_i}(0) \int_U u_{x_i} \, dx \\ &= \sum_{i=1}^n L_{p_i}(0) \underbrace{\int_{\partial U} u \nu^i \, dS}_{u \text{ has zero trace.}} = 0 \end{aligned}$$

Therefore, the previous inequality becomes

$$\begin{aligned}
 \theta \int_U |Du|^2 dx &\leq C \|f\|_{L^2(U)} \|Du\|_{L^2(U)} \\
 \implies \theta \|Du\|_{L^2(U)}^2 &\leq C \|f\|_{L^2(U)} \|Du\|_{L^2(U)} \\
 \implies \theta \|Du\|_{L^2(U)} &\leq C \|f\|_{L^2(U)} \\
 \stackrel{\text{Poincaré}}{\implies} \|u\|_{L^2(U)} &\leq C \|f\|_{L^2(U)}
 \end{aligned}$$

□

1.5 Constraints

In many optimisation problems, we are not free to choose an arbitrary point. We may want to minimise or maximise a function subject to some additional condition. For example, in multivariable calculus we might want to find the extreme values of $f(x, y)$ while restricting (x, y) to a curve $g(x, y) = k$. The constraint prevents us from moving in every direction, and at a constrained extremum the level curve of f becomes tangent to the constraint curve. This is the geometric idea behind the method of Lagrange multipliers.

A similar situation occurs in the calculus of variations. In this case, we optimise a functional over admissible sets. Without any restrictions, a minimiser can be varied in any admissible direction. When a constraint is imposed, however, only certain functions are allowed, and the minimiser must be found within this smaller class.

There are several ways in which such constraints can arise. And one of them is integral constraint:

1.5.1 Nonlinear eigenvalue problems

Now, we want to minimize the energy functional

$$I[w] = \frac{1}{2} \int_U |Dw|^2 dx \tag{1.77}$$

with $w = 0$ on ∂U and subjected to the constraint

$$J[w] = \int_U G(w) \, dx = 0, \quad (1.78)$$

where we are given $G : \mathbb{R} \rightarrow \mathbb{R}$ is a smooth function.

Let $g = G'$, and assume that

$$\begin{cases} |g(z)| \leq C(1 + |z|), \\ |G(z)| \leq C(1 + |z|^2) \end{cases}, \quad z \in \mathbb{R} \quad (1.79)$$

for some constant C .

We also introduce a new admissible class

$$\mathcal{A} = \{w \in H_0^1(U) \mid J[w] = 0\}$$

We also suppose that U is open, bounded, connected with smooth boundary.

Theorem 1.5.1 (Existence of constrained minimizer).

Assume that set $\mathcal{A}$ is nonempty. Then there exists $u \in \mathcal{A}$ satisfying

$$I[u] = \min_{w \in \mathcal{A}} I[w]$$

Proof. Let $\{u_k\}_{k=1}^\infty \subset \mathcal{A}$ be a minimizing sequence such that

$$I[u_k] \rightarrow m = \inf_{w \in \mathcal{A}} I[w]$$

Like previously, we can find a subsequence $\{u_{k_j}\}$ such that

$$u_{k_j} \rightarrow u \text{ weakly in } H_0^1(U) \quad (1.80)$$

and $I[u] \leq m$.

From compactness theorem, we get

$$u_{k_j} \rightarrow u \text{ in } L^2(U) \quad (1.81)$$

Claim:

$$J[u] = 0 \quad (1.82)$$

We see that,

$$\begin{aligned}
 |J[u]| &\stackrel{u_k \in \mathcal{A}}{=} |J[u] - J[u_k]| \\
 &\leq \int_U |G(u) - G(u_k)| \, dx \stackrel{M.V.T}{=} \int_U |G'(u)| |u - u_k| \, dx \\
 &\stackrel{1.79}{\leq} C \int_U |u - u_k| (1 + |u| + |u_k|) \, dx \\
 &\rightarrow 0 \text{ as } k \rightarrow \infty
 \end{aligned}$$

□

Theorem 1.5.2 (Lagrange multiplier).

Let $u \in \mathcal{A}$ satisfying

$$I[u] = \min_{w \in \mathcal{A}} I[w] \quad (1.83)$$

Then there exists $\lambda \in \mathbb{R}$ such that

$$\int_U Du \cdot Dv \, dx = \lambda \int_U g(u)v \, dx, \quad \forall v \in H_0^1(U) \quad (1.84)$$

Remark 1.5.3. fdfdfd

Proof. Let $v \in H_0^1(U)$. Assume that

$$g(u) \text{ is not equal to zero a.e. within } U. \quad (1.85)$$

Let $w \in H_0^1(U)$ such that

$$\int_U g(u)w \, dx \neq 0 \quad (1.86)$$

Now, define the function

$$j(\tau, \sigma) = J[u + \tau v + \sigma w] = \int_U G(u + \tau v + \sigma w) \, dx, \quad \tau, \sigma \in \mathbb{R} \quad (1.87)$$

Since $u \in \mathcal{A}$, we observe that

$$j(0, 0) = \int_U G(u) \, dx = 0 \quad (1.88)$$

Since j is C^1 , we have

$$\frac{\partial j}{\partial \tau}(\tau, \sigma) = \int_U g(u + \tau v + \sigma w) v \, dx \quad (1.89)$$

$$\frac{\partial j}{\partial \sigma}(\tau, \sigma) = \int_U g(u + \tau v + \sigma w) w \, dx \quad (1.90)$$

1.86 implies that

$$\frac{\partial j}{\partial \sigma}(0, 0) \neq 0 \quad (1.91)$$

Implicit Function Theorem implies that there exists a C^1 function $\phi : \mathbb{R} \rightarrow \mathbb{R}$ such that

$$\phi(0) = 0 \quad (1.92)$$

and

$$j(\tau, \phi(\tau)) = 0 \quad (1.93)$$

for sufficiently small τ .

Differentiating 1.93, we get

$$\frac{\partial j}{\partial \tau}(\tau, \phi(\tau)) + \frac{\partial j}{\partial \sigma}(\tau, \phi(\tau))(\phi'(\tau)) = 0 \quad (1.94)$$

1.89, 1.90, and 1.94 imply that

$$\phi'(0) = -\frac{\int_U g(u) v \, dx}{\int_U g(u) w \, dx} \quad (1.95)$$

Now, define

$$i(\tau) = I[u + \tau v + \phi(\tau)w]$$

Observe that i is C^1 and 1.93 implies that $J[u + \tau v + \phi(\tau)w] = 0$, so $u + \tau v + \phi(\tau)w \in \mathcal{A}$. So, i has a minimum at 0.

$$\begin{aligned}
0 = i'(0) &= \int_U (Du + \tau Dv + \phi(\tau) Dw) \cdot (Dv + \phi'(\tau) Dw) \, dx \Big|_{\tau=0} \\
&= \int_U Du \cdot (Dv + \phi'(0) Dw) \, dx \\
&= \int_U Du \cdot Dv \, dx + \phi'(0) \int_U Du \cdot Dw \, dx \\
\implies \int_U Du \cdot Dv \, dx &= -\phi'(0) \int_U Du \cdot Dw \, dx \\
\stackrel{1.95}{\implies} \int_U Du \cdot Dv \, dx &= \left(\frac{\int_U g(u)v \, dx}{\int_U g(u)w \, dx} \right) \int_U Du \cdot Dw \, dx \\
&\quad \text{define } \lambda = \frac{\int_U Du \cdot Dw \, dx}{\int_U g(u)w \, dx} \\
\int_U Du \cdot Dv \, dx &= \lambda \int_U g(u)v \, dx, \quad \forall v \in H_0^1(U)
\end{aligned}$$

Now, suppose that

$$g(u) = 0 \quad a.e. \text{ in } U$$

Approximating g by bounded functions, we deduce $DG(u) = g(u)Du = 0$ a.e. Since, by assumption, U is connected, $G(u)$ is constant a.e. It follows that $G(u) = 0$ a.e. since for $u \in \mathcal{A}$, $J[u] = \int_U G(u) \, dx = 0$. As $u = 0$ on ∂U in the trace sense, it follows that $G(0) = 0$

This implies $0 \in \mathcal{A}$. And since u is a minimizer, we have $I[u] \leq I[0] = 0$. But $I[u] = 1/2 \int_U |Du|^2 \, dx \geq 0$.

$$0 \leq I[u] \leq 0 \implies I[u] = 0$$

$$\implies Du = 0 \text{ a.e.}$$

since U is connected, u is a constant a.e.

since $u \in H_0^1(U)$, $u = 0$ a.e.

Since, $g(u) = 0$ a.e., the identity 1.84 is trivially valid in this case, for any λ .

Reason:

- (i) In Sobolev spaces, the chain rule says, if $u \in W^{1,p}(U)$ and $f \in C^1(\mathbb{R})$ such that $f' \in L^\infty(\mathbb{R})$ and $f(0) = 0$, then $f \circ u \in W^{1,p}(U)$ and

$$D_j(f \circ u) = f'(u)D_j u \text{ a.e. in } U.$$

- (ii) on ∂U , $u = 0$, and $G(u) = G(0)$. We can't use the argument that $G(u) = 0$ a.e. and use this to extend to the boundary ∂U since both the boundary and the point 0 have measure zero. To prove this, we use the method of contradiction.

Suppose $G(0) \neq 0$.

Since G is continuous, $\exists \delta$ s.t. $G(z) \neq 0$ for $|z| < \delta$.

But $G(u) = 0$ a.e. in U . This implies the set $\{x \in U \mid |u(x)| \leq \delta\}$ has measure zero. This follows that $|u(x)| > \delta$ a.e. in U . We observe that if $u \in H_0^1(U)$, then so is $|u| \in H_0^1(U)$. But from Trace theory, $|u| = 0$ on ∂U , a zero Lebesgue measure set. Thus, a contradiction.

Therefore, $G(0) = 0$. See [this Math StackExchange discussion](#).

□

1.5.2 Unilateral constraints

Theorem 1.5.4 (Existence of minimizer).

Assume the admissible set $\mathcal{A}$ is nonempty. Then there exists a unique function $u \in \mathcal{A}$ satisfying

$$I[u] = \min_{w \in \mathcal{A}} I[w]$$

Proof. Like before, we take a minimizing sequence $\{u_k\}_{k=1}^\infty \subset \mathcal{A}$ satisfying

$$I[u_k] \rightarrow m = \inf_{w \in \mathcal{A}} I[w] \tag{1.96}$$

We can find a subsequence $\{u_{k_j}\}$ such that

$$u_{k_j} \rightarrow u \quad \text{weakly in } H_0^1(U) \tag{1.97}$$

And, by compactness theorem,

$$u_{k_j} \rightarrow u \quad \text{in } L^2(U) \quad (1.98)$$

We have $u_{k_j} \geq h.a.e.$ This implies $u \geq h.a.e.$ Therefore $u \in \mathcal{A}$.

To prove Uniqueness, we use the method of contradiction. We assume u and $\tilde{u} \in \mathcal{A}$ are two minimizers of I and $u \neq \tilde{u}$. Since $\mathcal{A}$ is convex, we can define $w = \frac{u+\tilde{u}}{2} \in \mathcal{A}$.

Now,

$$I[w] = \int_U \frac{1}{2} \left| \left(\frac{Du + D\tilde{u}}{2} \right) \right|^2 - f\left(\frac{u + \tilde{u}}{2}\right) dx$$

=

□

Theorem 1.5.5 (Variational characterization of minimizer).

Let $u \in \mathcal{A}$ be a unique solution of

$$I[u] = \min_{w \in \mathcal{A}} I[w]$$

Then

$$\int_U Du \cdot D(w - u) dx \geq \int_U f(w - u) dx, \quad \forall w \in \mathcal{A} \quad (1.99)$$

1.99 is called variational inequality.

Proof. Fix $w \in \mathcal{A}$. Since $\mathcal{A}$ is convex, for each $0 \leq \tau \leq 1$,

$$u = \tau(w - u) = (1 - \tau)u + \tau w \in \mathcal{A}$$

Set

$$i(\tau) = I[u = \tau(w - u)]$$

Since u is the minimizer, $i(0) \leq i(\tau)$ for all $0 \leq \tau \leq 1$. Hence,

$$i'(0) \geq 0. \quad (1.100)$$

Now for $0 < \tau \leq 1$,

$$\begin{aligned}
\frac{i(\tau) - i(0)}{\tau} &= \frac{1}{\tau} \int_U \frac{|Du + \tau D(w - u)|^2 - |Du|^2}{2} - f(u\tau(w - u) - u) \, dx \\
&= \frac{1}{\tau} \int_U \frac{|Du|^2 + \tau^2 |D(w - u)|^2 + 2\tau Du \cdot D(w - u) - |Du|^2}{2} - \tau f(w - u) \, dx \\
&= \int_U \frac{\tau |D(w - u)|^2 + 2Du \cdot D(w - u)}{2} - f(w - u) \, dx
\end{aligned}$$

1.100 implies that

$$0 \leq i'(0) = \int_U Du \cdot D(w - u) - f(w - u) \, dx$$

□

1.5.3 Variational inequality interpretation

1.6 Critical Points

1.6.1 Deformation Theorem

Let H denote a real Hilbert space, with norm $\|\cdot\|$ and inner product $(\cdot, \cdot)$. Let $I : H \rightarrow \mathbb{R}$ be a nonlinear functional on H .

Definition 1.6.1. I is differentiable at $u \in H$ if there exists $v \in H$ such that

$$I[w] = I[u] + (v, w - u) + o(\|w - u\|), \quad w \in H \quad (1.101)$$

The function v is unique, and we write $I'[u] = v$

Definition 1.6.2. $I \in C^1(H; \mathbb{R})$ if I is differentiable for each $u \in H$ and the mapping $I' : H \rightarrow H$ is continuous.

We also assume that

$$I' : H \rightarrow H \text{ is Lipschitz continuous on bounded subsets of } H. \quad (1.102)$$

Notation 1.6.3.

(i) $\mathcal{C}$ denotes the collection of functionals $I \in C^1(H; \mathbb{R})$ satisfying 1.102.

(ii) Let $c \in \mathbb{R}$, then we define the sets

$$A_c = \{u \in H \mid I[u] \leq c\}, \quad K_c = \{u \in H \mid I[u] = c, I'[u] = 0\}$$

Definition 1.6.4.

(i) We define the function $u \in H$ as a critical point if $I'[u] = 0$.

(ii) $c \in \mathbb{R}$ is a critical value if $K_c \neq \emptyset$.

Definition 1.6.5. A functional $I \in C^1(H; \mathbb{R})$ satisfies the *Palais–Smale compactness condition* if each sequence $\{u_k\}_{k=1}^\infty \subset H$ such that

(i) $\{I[u_k]\}_{k=1}^\infty$ is bounded

(ii) $I'[u_k] \rightarrow 0$ in H^*

is precompact in H .

In the next theorem, we will see that if c is not a critical value, then we can continuously push every function from $A_{c+\delta}$ down into the set $A_{c-\delta}$.

Theorem 1.6.6 (Deformation Theorem).

Assume $I \in \mathcal{C}$ satisfies the Palais – Smale condition. Suppose

$$K_c = \emptyset \tag{1.103}$$

Then for each sufficiently small $\epsilon > 0$, there exists a constant $0 < \delta < \epsilon$ and a function

$$\eta \in C([0, 1] \times H; H)$$

defined by

$$\eta_t(u) = \eta(t, u), \quad (0 \leq t \leq 1, u \in H)$$

satisfying

(i) $\eta_0(u) = u, \quad u \in H$

(ii) $\eta_1(u) = u, \quad u \notin I^{-1}[c - \epsilon, c + \epsilon]$

$$(iii) \quad I[\eta_t(u)] \leq I[u], \quad u \in \mathbb{R}, 0 \leq t \leq 1$$

$$(iv) \quad \eta_1(A_{c+\delta}) \subset A_{c-\delta}$$

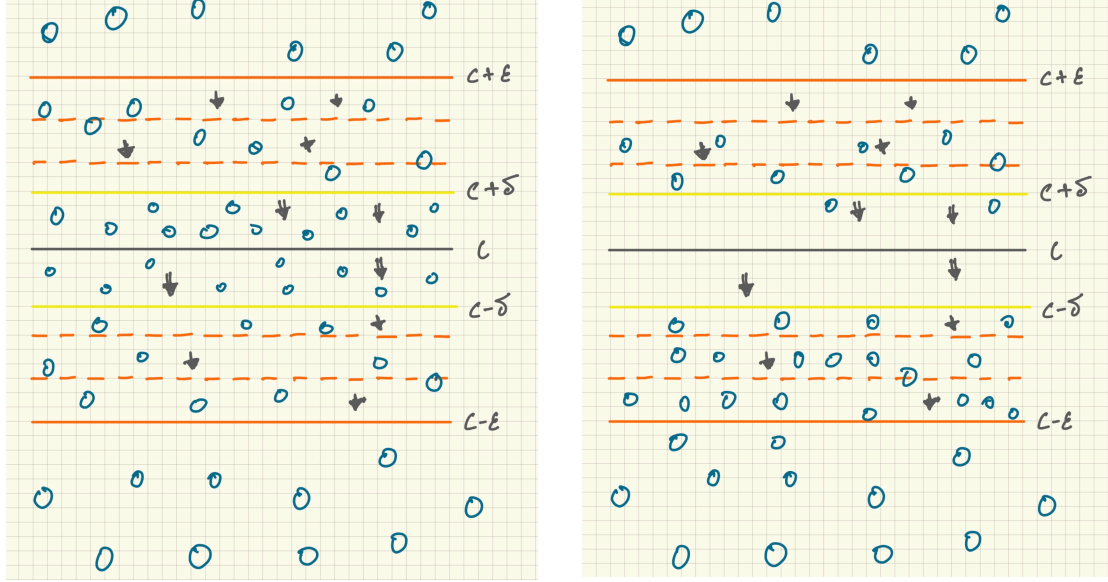


Figure 1.6.1: Points with energy levels in the range $c - \varepsilon \leq I[u] \leq c + \varepsilon$ are pushed down, with points in the smaller band $c - \delta \leq I[u] \leq c + \delta$ being pushed down more strongly. Points outside the ε -band are unaffected. A point between $c + \delta$ and $c + \varepsilon$ may be pushed below $c + \delta$ during the deformation.

idea of the proof: We first construct a downhill vector field that pushes points near the level c towards lower energy, while leaving points far from c unchanged. We then solve an ODE to obtain a continuous mapping, which gives the desired deformation. Then we show that the energy of deformed functions around c is less than the original points.

Proof. Fix δ to satisfy

$$0 < \delta < \varepsilon, \quad 0 < \delta < \frac{\sigma^2}{2} \quad (1.104)$$

Set

$$\begin{aligned} A &= \{u \in H \mid I[u] \leq c - \varepsilon \text{ or } I[u] \geq c + \varepsilon\} \\ B &= \{u \in H \mid c - \delta \leq I[u] \leq c + \delta\} \end{aligned}$$

From eqn 1.102, I' is Lipschitz continuous on bounded subsets of H .

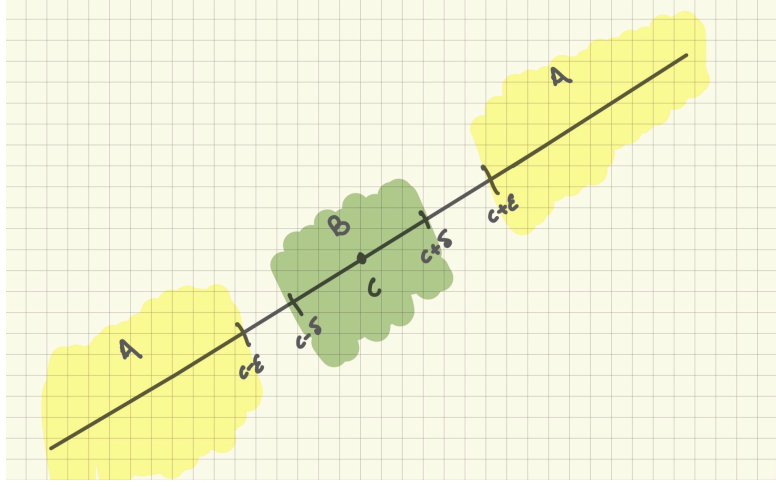


Figure 1.6.2: A and B are clearly disjoint.

FACT: $u \mapsto \text{dist}(u, S)$ is 1-Lipschitz for any set $S \subset H$. Now, consider the mapping $u \mapsto \text{dist}(u, A) + \text{dist}(u, B)$. For any $u \in H$, triangle inequality gives

$$\text{dist}(u, A) + \text{dist}(u, B) \geq \text{dist}(A, B) > 0, \text{ as both } A \text{ and } B \text{ are clearly disjoint.}$$

Therefore, the mapping $u \mapsto \text{dist}(u, A) + \text{dist}(u, B)$ is bounded below by a positive constant on each bounded subset of H .

Define the function

$$g(u) = \frac{\text{dist}(u, A)}{\text{dist}(u, A) + \text{dist}(u, B)}, \quad u \in H$$

g is Lipschitz on bounded sets, and satisfies

$$0 \leq g \leq 1, \quad g = 0 \text{ on } A, \quad g = 1 \text{ on } B. \quad (1.105)$$

Set

$$h(t) = \begin{cases} 1, & 0 \leq t \leq 1 \\ 1/t, & t \geq 1 \end{cases} \quad (1.106)$$

Define the mapping $V : H \mapsto H$ by

$$V(u) = -g(u)h(\|I'[u]\|)I'[u], \quad u \in H \quad (1.107)$$

Observe that

$$\begin{aligned}\|V(u)\| &= \|g(u)\| \|h(\|I'[u]\|)\| \|I'[u]\| \\ &\leq \|h(\|I'[u]\|)\| \|I'[u]\|\end{aligned}$$

If $\|I'[u]\| \leq 1$, then $\|h(\|I'[u]\|)\| = 1$

If $\|I'[u]\| \geq 1$, then $\|h(\|I'[u]\|)\| = \frac{1}{\|I'[u]\|}$

So, $\|V(u)\| \leq 1$. V is bounded.

Consider now for each $u \in H$ the ODE

$$\begin{cases} \frac{d}{dt}\eta(t) = V(\eta(t)), & t > 0 \\ \eta(0) = u \end{cases} \quad (1.108)$$

As V is bounded and Lipschitz continuous on bounded sets, there exists a unique solution $\eta = \eta(t, u) = \eta_t(u)$ for all $t \geq 0$. Restricting to $0 \leq t \leq 1$, the mapping $\eta \in C([0, 1] \times H; H)$ satisfies the assertion (i). For assertion (ii), we observe that, for $u \notin A$,

$$\begin{aligned}g(u) &= 0 \\ \implies V(u) &= 0 \\ \implies \frac{d}{dt}\eta_t(u) &= 0 \\ \implies \eta_t(u) &= u, \quad \forall t \geq 0. \\ \implies \eta_1(u) &= u, \quad u \notin I^{-1}[c - \epsilon, c + \epsilon]\end{aligned}$$

Now computing

$$\begin{aligned}\frac{d}{dt}I[\eta_t(u)] &= \left(I'[\eta_t(u)], \frac{d}{dt}\eta_t(u) \right) \\ &= (I'[\eta_t(u)], V(\eta_t(u))) \\ &= -g(\eta_t(u)) h(\|I'[\eta_t(u)]\|) \|I'[\eta_t(u)]\|^2 \\ &\leq 0, \quad u \in H, \quad 0 \leq t \leq 1.\end{aligned} \quad (1.109)$$

Now, integrating the above inequality from 0 to t ,

$$\begin{aligned} I[\eta_t(u)] - I[\eta_0(u)] &\leq 0 \\ \implies I[\eta_t(u)] &\leq I[u] \end{aligned}$$

So, assertion (iii) is valid.

Fix any point

$$u \in A_{c+\delta} \tag{1.110}$$

We want to prove

$$\eta_1(u) \in A_{c-\delta} \tag{1.111}$$

and verify assertion (iv).

Claim: There exists constants $0 < \sigma, \epsilon < 1$ such that

$$\|I'[u]\| \geq \sigma \text{ for each } u \in A_{c+\epsilon} - A_{c-\epsilon} \tag{1.112}$$

We will use the method of contradiction. Suppose 1.112 is false for all $\sigma, \epsilon > 0$, there exists sequences $\sigma_k \rightarrow 0, \epsilon_k \rightarrow 0$, and elements

$$u_k \in A_{c+\epsilon} - A_{c-\epsilon} \tag{1.113}$$

with

$$\|I'[u_k]\| \leq \sigma_k \tag{1.114}$$

Since I satisfies the Palais-Smale condition, there exists a subsequence $\{u_{k_j}\}_{j=1}^\infty$ and $u \in H$ with $u_{k_j} \rightarrow u$ in H . Since $I \in \mathbb{C}^1(H; \mathbb{R})$, as $\epsilon \rightarrow 0$, 1.113 and 1.114 imply $I[u] = c, I'[u] = 0$. So, $K_c \neq \emptyset$. A contradiction.

Now, if $\eta_t(u) \notin B$ for some $0 \leq t \leq 1$, then either $I[\eta_t(u)] < c - \delta$ or $I[\eta_t(u)] > c + \delta$. Since we have taken $u \in A_{c+\delta}$, only option is $I[\eta_t(u)] < c - \delta$. Therefore, $\eta_1(u) \in A_{c-\delta}$.

Now, suppose $\eta_t(u) \in B$, for $0 \leq t \leq 1$. Then $g(\eta_t(u)) = 1$. 1.109 gives

$$\frac{d}{dt} I[\eta_t(u)] = -h(\|I'[\eta_t(u)]\|) \|I'[\eta_t(u)]\|^2 \tag{1.115}$$

If $\|I'[\eta_t(u)]\| \leq 1$, then

$$\frac{d}{dt}I[\eta_t(u)] = -\|I'[\eta_t(u)]\|^2 \stackrel{1.112}{\leq} -\sigma^2$$

If $\|I'[\eta_t(u)]\| \geq 1$, then

$$\frac{d}{dt}I[\eta_t(u)] = -\|I'[\eta_t(u)]\| \stackrel{1.112}{\leq} -\sigma \leq -\sigma^2$$

So, we have

$$\frac{d}{dt}I[\eta_t(u)] \leq -\sigma^2$$

Integrating the above inequality from 0 to 1,

$$\begin{aligned} I[\eta_1(u)] - I[\eta_0(u)] &\leq -\sigma^2 \\ \implies I[\eta_1(u)] &\leq I[\eta_0(u)] - \sigma^2 \stackrel{1.110}{\leq} c + \delta - \sigma^2 \stackrel{1.104}{\leq} c - \delta \end{aligned}$$

This establishes 1.111 and completes the proof. □

1.6.2 Mountain Pass Theorem

Theorem 1.6.7 (Mountain Pass Theorem).

Assume that the functional $I \in \mathcal{C}$ satisfies the Palais – Smale condition. Suppose that

- (i) $I[0] = 0$,
- (ii) *there exist constants $r, a > 0$ such that*

$$I[a] \geq a \text{ if } \|u\| = r$$

- (iii) *and there exists $v \in H$ with*

$$\|v\| > r, I[v] \leq 0$$

Define

$$\Gamma = \{g \in C([0, 1]; H) \mid g(0) = 0, g(1) = v\}$$

Then

$$c = \inf_{g \in \Gamma} \max_{0 \leq t \leq 1} I[g(t)]$$

is a critical value of I .

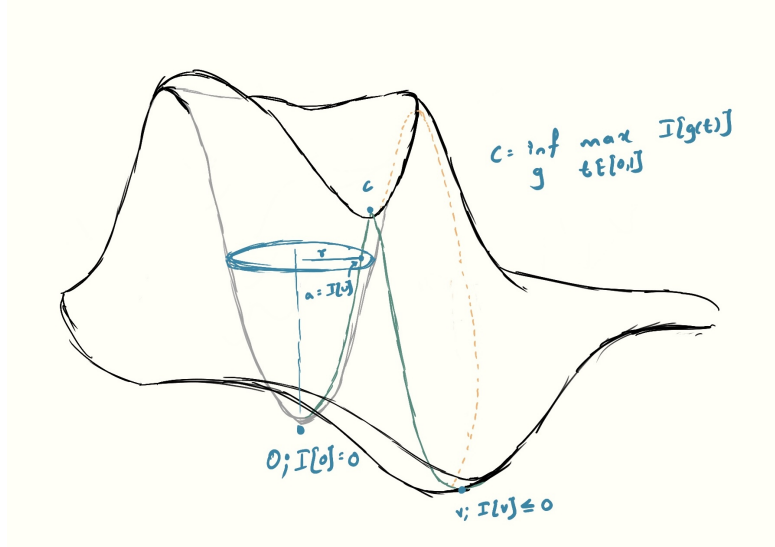


Figure 1.6.3: Mountain Pass Theorem visualisation.

idea of the proof: We assume c is not a critical value, so we can apply the Deformation Theorem, which deforms the $A_{c+\delta}$ set into the $A_{c-\delta}$ set. Now we choose a map such that its highest energy level is less than or equal to $c + \delta$. We then deform it, so its energy is now less than or equal to $c - \delta$. But according to the definition of c , no admissible path can have its highest energy level below c , since

$$c = \inf_{g \in \Gamma} \max_{0 \leq t \leq 1} I(g(t)).$$

So we have found an admissible path whose highest energy level is at most $c - \delta < c$, which is a contradiction.

Proof. Clearly

$$c \geq a \tag{1.116}$$

Suppose c is not a critical value of I such that

$$K_c = \emptyset \tag{1.117}$$

Let ϵ be a sufficiently small number such that

$$0 < \epsilon < \frac{a}{2} \quad (1.118)$$

From Deformation Theorem 1.6.6, there exist a constant $0 < \delta < \epsilon$ and function $\eta : H \rightarrow H$ with

$$\eta(A_{c+\delta}) \subset \eta(A_{c-\delta}) \quad (1.119)$$

and

$$\eta(u) = u \quad \text{if } u \notin I^{-1}[c - \delta, c + \delta] \quad (1.120)$$

Let $g \in \Gamma$ satisfying

$$\max_{0 \leq t \leq 1} I[g(t)] \leq c + \delta \quad (1.121)$$

We note that $\eta(g(0)) = \eta(0) = 0$, and $\eta(g(1)) = \eta(v) = v$ by 1.120 Therefore, $\tilde{g} = \eta \circ g \in \Gamma$

But equation 1.121 and Deformation Theorem imply

$$\max_{0 \leq t \leq 1} I[\tilde{g}] \leq c - \delta \quad (1.122)$$

And, from our assumption,

$$c = \inf_{g \in \Gamma} \max_{0 \leq t \leq 1} I[g(t)] \stackrel{1.122}{\leq} c - \delta$$

A contradiction. □

1.6.3 Application to semilinear elliptic PDE

Now we investigate the semilinear BVP:

$$\begin{cases} -\Delta u = f(u), & \text{in } U \\ u = 0, & \text{on } \partial U \end{cases} \quad (1.123)$$

Assume f is smooth, and for some

$$1 < p < \frac{n+2}{n-2} \quad (1.124)$$

we have

$$|f(z)| \leq C(1 + |z|^p), \quad |f'(z)| \leq C(1 + |z|^{p-1}), \quad z \in \mathbb{R} \quad (1.125)$$

where C is constant. Also, assume that

$$0 \leq F(z) \leq \gamma f(z)z, \quad \text{for some constant } \gamma < 1/2, \quad (1.126)$$

where $F(z) = \int_0^z f(s) ds$, and $z \in \mathbb{R}$. We also suppose that for constants $0 < a \leq A$

$$a|z|^{p+1} \leq |F(z)| \leq A|z|^{p+1}, \quad z \in \mathbb{R} \quad (1.127)$$

We observe that $u \equiv 0$ is a trivial solution, since from 1.127, $f(0) = 0$

We would like to investigate if the above PDE has any non trivial solution.

Theorem 1.6.8 (Existence).

The BVP 1.123 has at least one weak solution $u \not\equiv 0$.

Proof. Define

$$I[u] = \int_U \frac{1}{2} |Du|^2 - F(u) dx, \quad u \in H_0^1(U). \quad (1.128)$$

Goal: to apply Mountain Pass Theorem.

We recall the norma and inner product for the space $H = H_0^1(U)$.

$$\|u\| = (\int_U |Du|^2 dx)^{1/2}, \quad (u, v) = \int_U Du \cdot Dv dx, \quad u, v \in H.$$

Then,

$$I[u] = \frac{1}{2} \|u\|^2 - \int_U F(u) dx = I_1[u] - I_2[u]$$

Claim:

$$I \in \mathcal{C}. \quad (1.129)$$

Note that for $u, w \in H$,

$$I_1[w] = \frac{1}{2} \|w\|^2 = \frac{1}{2} \|u + w - u\|^2 = \frac{1}{2} \|u\|^2 + (u, w - u) + \frac{1}{2} \|w - u\|^2$$

Therefore, I_1 is differentiable at u , with $I_1'[u] = u$. Hence, $I_1 \in \mathcal{C}$.

To examine I_2 , we recall Lax-Milgram Theorem:

Theorem 1.6.9 (Lax-Milgram Theorem).

for every $v^* \in H^{-1}(U)$, there exists a unique $v \in H_0^1(U)$ for the BVP

$$\begin{cases} -\Delta v = v^* & \text{in } U \\ v = 0, & \text{on } U \end{cases}$$

such that

$$\int_U Dv \cdot D\phi = \langle v^*, \phi \rangle, \quad \phi \in H_0^1(U)$$

Define the operator $K : H^{-1}(U) \rightarrow H_0^1(U)$ such that

$$K(v^*) = v, \quad K = (-\Delta)^{-1}$$

Claim: K is an isometry, $\|Kv^*\|_{H_0^1(U)} = \|v^*\|_{H^{-1}(U)}$

Recall

$$\|v^*\|_{H^{-1}(U)} = \sup_{\phi \neq 0 \in H_0^1(U)} \frac{|\langle v^*, \phi \rangle|}{\|\phi\|_{H_0^1(U)}}$$

We know,

$$\begin{aligned} |\langle v^*, \phi \rangle| &= |(v, \phi)| \stackrel{\text{Cauchy-Schwartz}}{\leq} \|v\| \|\phi\| \\ \implies \frac{|\langle v^*, \phi \rangle|}{\|\phi\|} &\leq \|v\| \\ \implies \sup_{\phi \neq 0} \frac{|\langle v^*, \phi \rangle|}{\|\phi\|} &\leq \|v\| \\ \implies \|v^*\|_{H^{-1}(U)} &\leq \|v\| = \|Kv^*\|_{H_0^1(U)} \end{aligned}$$

For the other inequality, take the test function $\phi = v$. This implies

$$\begin{aligned} (v, v) &= \langle v^*, v \rangle \\ \implies \|v\|^2 &= |\langle v^*, v \rangle| \leq \|v^*\|_{H^{-1}(U)} \|v\| \\ \implies \|v\| &\leq \|v^*\|_{H^{-1}(U)} \\ \implies \|Kv^*\|_{H_0^1(U)} &\leq \|v^*\|_{H^{-1}(U)} \end{aligned}$$

Hence, K is an isometry.

Claim: $f(u) \in H^{-1}(U)$.

Define the functional

$$F_{f(u)} : H_0^1(U) \rightarrow \mathbb{R}, \quad F_{f(u)}(\phi) = \int_U f(u)\phi \, dx.$$

Clearly, $F_{f(u)}$ is linear in ϕ .

To check that $F_{f(u)}$ is bounded, let

$$q = \frac{2n}{n+2}, \quad q' = \frac{2n}{n-2} = 2^*, \text{ the Sobolev conjugate of } 2$$

Also,

$$(2^*)' = \frac{2n}{n+2} = q.$$

now, we show that

$$f(u) \in L^q(U).$$

Since, by imbedding theorems

$$u \in H_0^1(U) \hookrightarrow L^{2^*}(U),$$

we have

$$\int_U |u|^{2^*} \, dx < \infty.$$

By the growth condition [1.125](#),

$$|f(u(x))| \leq C(1 + |u(x)|^p).$$

Therefore,

$$|f(u)|^q \leq C(1 + |u|^p)^q \leq C(1 + |u|^{pq}).$$

Since

$$p < \frac{n+2}{n-2},$$

we have

$$pq = p \frac{2n}{n+2} < \frac{n+2}{n-2} \frac{2n}{n+2} = \frac{2n}{n-2} = 2^*.$$

Thus, since U is bounded and $u \in L^{2^*}(U)$,

$$u \in L^{pq}(U).$$

Consequently,

$$\int_U |f(u)|^q dx \leq C \left(|U| + \int_U |u|^{pq} dx \right) < \infty.$$

Hence

$$f(u) \in L^q(U) = L^{\frac{2n}{n+2}}(U).$$

Now, for $\phi \in H_0^1(U)$, Hölder's inequality gives

$$\begin{aligned} |F_{f(u)}(\phi)| &= \left| \int_U f(u) \phi dx \right| \\ &\leq \|f(u)\|_q \|\phi\|_{q'} \\ &= \|f(u)\|_{\frac{2n}{n+2}} \|\phi\|_{2^*}. \end{aligned}$$

By the Sobolev imbedding inequality,

$$\|\phi\|_{2^*} \leq C \|\phi\|_{H_0^1(U)}.$$

Therefore,

$$|F_{f(u)}(\phi)| \leq C \|f(u)\|_{\frac{2n}{n+2}} \|\phi\|_{H_0^1(U)}.$$

Thus $F_{f(u)}$ is a bounded linear functional on $H_0^1(U)$. Hence

$$F_{f(u)} \in (H_0^1(U))^* = H^{-1}(U).$$

Therefore, identifying $f(u)$ with the corresponding functional,

$$f(u) \in H^{-1}(U)$$

Claim: If $u \in H_0^1(U)$, then

$$I_2'[u] = K[f(u)]. \tag{1.130}$$

To see this, we recall the Taylor's theorem for integral remainder:

$$F(a+b) = F(a) + f(a)b + \int_0^1 (1-s)f'(a+sb) ds b^2$$

Thus, for each $w \in H_0^1(U)$,

$$\begin{aligned} I_2[w] &= \int_U F(w) dx = \int_U F(u+w-u) dx \\ &= \int_U F(u) + f(u)(w-u) dx + R \\ &= I_2(u) + \int_U DK[f(u)] \cdot D(w-u) dx + R, \end{aligned} \tag{1.131}$$

where the remainder term R satisfies, according to 1.125,

$$\begin{aligned} |R| &\leq C \int_U (1 + |u|^{p-1} + |w-u|^{p-1}) |w-u|^2 dx \\ &\leq c \left(\int_U |w-u|^2 + |w-u|^{p+1} dx \right) \\ &\quad + C \left(\int_U |u|^{p+1} dx \right)^{\frac{p-1}{p+1}} \left(\int_U |w-u|^{p+1} dx \right)^{\frac{2}{p+1}} \end{aligned}$$

Since $p+1 < 2^*$, the Sobolev inequalities show $R = o(\|w-u\|)$.

Since K is an isometry, we see that

$$I_2[w] = I_2[u] + (K[f(u)], w-u) + o(\|w-u\|).$$

We note that if $u, \tilde{u} \in H_0^1(U)$ with $\|u\|, \|\tilde{u}\| \leq L$, then

$$\begin{aligned} \|I_2'[u] - I_2'[\tilde{u}]\| &= \|K[f(u)] - K[f(\tilde{u})]\|_{H_0^1(U)} \\ &\stackrel{K \text{ is an isometry}}{=} \|f(u) - f(\tilde{u})\|_{H^{-1}(U)} \\ &\leq \|f(u) - f(\tilde{u})\|_{L^{\frac{2n}{n+2}}(U)} \end{aligned}$$

But

$$\begin{aligned}
& \|f(u) - f(\tilde{u})\|_{L^{\frac{2n}{n+2}}(U)} \\
& \stackrel{\substack{1.125 \\ M.V.T}}{\leq} C \left(\int_U [(1 + |u|^{p-1} + |\tilde{u}|^{p-1}) |u - \tilde{u}|^{\frac{2n}{n+2}}] dx \right)^{\frac{n+2}{2n}} \\
& \stackrel{\substack{\text{Hölder, } r=\frac{n+2}{4} \\ s=\frac{n+2}{n-1}}}{\leq} \left(\int_U \left[(1 + |u|^{p-1} + |\tilde{u}|^{p-1}) |u - \tilde{u}|^{\frac{2n}{n+2} * \frac{n+2}{4}} \right] dx \right)^{2/n} \|u - \tilde{u}\|_{L^{2^*}(U)} \\
& \leq C(L) \|u - \tilde{u}\|_{L^{2^*}(U)} \stackrel{\text{Sobolev imbedding}}{\leq} C(L) \|u - \tilde{u}\|_{H_0^1(U)}
\end{aligned}$$

Thus, $I'_2 : H_0^1(U) \rightarrow H_0^1(U)$ is Lipschitz continuous on bounded sets. Hence, $I_2 \in \mathcal{C}$, and we have established assertion 1.129.

Verifying the Palais-Smale condition:

Let $\{u_k\}_{k=1}^\infty \subset H_0^1(U)$ satisfying

$$\{I[u_k]\}_{k=1}^\infty \text{ bounded} \quad (1.132)$$

and

$$I'[u_k] \rightarrow 0 \quad \text{in } H_0^1(U). \quad (1.133)$$

From earlier, $I'[u_k] = u_k - K[f(u_k)]$ So

$$u_k - K(f(u_k)) \rightarrow 0, \quad \text{in } H_0^1(U) \quad (1.134)$$

Thus, for each $\epsilon > 0$,

$$\begin{aligned}
& |(I'[u_k], v)| = |(u_k - K[f(u_k)], v)| = |(u_k, v) - (K[f(u_k)], v)| \\
& = \left| \int_U Du_k \cdot Dv \, dx - \langle f(u_k), v \rangle \right|, \quad \text{by definition of } K, (K[f(u_k)], v) = \langle f(u_k), v \rangle \\
& = \left| \int_U Du_k \cdot Dv - f(u_k)v \, dx \right| \stackrel{\substack{\text{C.S inequality} \\ 1.133}}{\leq} \epsilon \|v\|, \quad v \in H_0^1(U)
\end{aligned}$$

for large enough k .

Let $v = u_k$, then

$$\left| \int_U |Du_k|^2 - f(u_k)u_k \, dx \right| \leq \epsilon \|u_k\|$$

for each $\epsilon > 0$ and for all k sufficiently large. For $\epsilon = 1$, we see that

$$\int_U f(u_k) u_k \, dx \leq \|u_k\|^2 + \|u_k\| \quad (1.135)$$

for all sufficiently large k .

But 1.132 implies

$$\left(\frac{1}{2} \|u_k\|^2 - \int_U F(u_k) \, dx \right) \leq C < \infty$$

for all k and constant C , we deduce that

$$\begin{aligned} \|u_k\|^2 &\leq C + 2 \int_U F(u_k) \, dx \\ &\stackrel{1.126}{\leq} C + 2\gamma \int_U f(u_k) u_k \, dx \\ &\stackrel{1.135}{\leq} C + 2\gamma(\|u_k\|^2 + \|u_k\|) \end{aligned}$$

Since, by assumption, $\gamma < 1$, $\{u_k\}_{k=1}^\infty$ is bounded in $H_0^1(U)$.

Hence, there exists a subsequence $\{u_{k_j}\}_{j=1}^\infty$ and $u \in H_0^1(U)$, with $u_{k_j} \rightharpoonup u$ weakly in $H_0^1(U)$. By compactness theorem, $u_{k_j} \rightarrow u$ in $L^{p+1}(U)$, since $(p+1) < 2^*$. And $\|f(u_k) - f(u)\| \leq C|(1 + \|u_k\|^{p-1} + \|u\|^{p-1})| \|u_k - u\|_{L^{p+1}(U)} \rightarrow 0$. Also, since K is continuous, we have $K[f(u_k)] \rightarrow K[f(u)]$ in $H_0^1(U)$.

Consequently, 1.134 implies

$$u_{k_j} \rightarrow u, \quad \text{in } H_0^1(U). \quad (1.136)$$

Verifying Mountain Pass Theorem:

$$I[0] = 1/2 \|0\|^2 - \int_U \int_0^0 f(s) \, ds = 0$$

Suppose now $u \in H_0^1(U)$, with $\|u\| = r > 0$.

Then

$$I[u] = I_1[u] - I_2[u] = \frac{r^2}{2} - I_2[u]. \quad (1.137)$$

Since $p+1 < 2^*$,

$$\begin{aligned}
|I_2[u]| &= \left| \int_U F(u) \, dx \right| \stackrel{1.127}{\leq} C \int_U |u|^{p+1} \, dx \\
r' &= \frac{2^*}{2^* - (p+1)} C \left(\int_U |u|^{2^*} \, dx \right)^{\frac{p+1}{2^*}} \\
&\stackrel{\text{H\"older, } r = \frac{2^*}{p+1}}{\leq} \\
&\stackrel{\text{Sobolev inequality}}{\leq} C \|u\|_{L^{2^*}(U)}^{p+1} \leq C \|u\|_{H_0^1(U)}^{p+1} = Cr^{p+1}
\end{aligned}$$

In view of 1.137,

$$I[u] \geq \frac{r^2}{2} - Cr^{p+1} \geq \frac{r^2}{4} = a > 0$$

provided $r > 0$ is small enough, since $p+1 > 2$. Now fix $u \in H, u \neq 0$. Let $v = tu$, for $t > 0$. Then

$$\begin{aligned}
I[v] &= I_1[tu] - I_2[tu] \\
&= t^2 I_1[u] - \int_U F(tu) \, dx \\
&\stackrel{1.127}{\leq} t^2 I_1[u] - at^{p+1} \int_U |u|^{p+1} \, dx \\
&< 0
\end{aligned}$$

for large enough $t > 0$.

By Mountain Pass Theorem, there exists a function $u \in H_0^1(U)$, with

$$I'[u] = u - K[f(u)] = 0.$$

In particular for each $v \in H_0^1(U)$,

$$\int_U Du \cdot Dv \, dx = \int_U f(u)c \, dx$$

Therefore, u is a weak solution of 1.123.

□

1.7 Some Exercise Solutions

Chapter 2

Nonvariational Techniques

In this chapter, we go through various techniques for proving the existence, non-existence, uniqueness and other properties of solutions for nonlinear elliptic partial differential equations that are not of variational form.

2.1 Monotonicity Methods

First of all, let us have a look at the BVP for a divergence quasilinear elliptic PDE:

$$\begin{cases} -\operatorname{div} \mathbf{a}(Du) = f & \text{in } U, \\ u = 0 & \text{on } \partial U. \end{cases} \quad (2.1)$$

where

- $f \in L^2(U)$ is given
- $\mathbf{a}$ is a smooth vector field; $\mathbf{a} : \mathbb{R}^n \rightarrow \mathbb{R}^n$, $\mathbf{a} = (a^1, \dots, a^n)$
- $u : \{U\} \rightarrow \mathbb{R}$ is the unknown.
- the domain U is open, bounded subset of $\mathbb{R}^n$ with smooth boundary.

Now, the question is "Can we apply the theory of Calculus of Variations" to this problem. Assume there exists a function $F : \mathbb{R}^n \rightarrow \mathbb{R}$ such that $\mathbf{a}$ is the gradient of F , i.e.

$$\mathbf{a}(p) = DF(p), \quad p \in \mathbb{R}^n, \quad (2.2)$$

Substituting this in equation 2.1, we have

$$\begin{aligned} -\operatorname{div}(\mathbf{a}(Du)) &= f \\ \implies -\operatorname{div}(DF(Du)) &= f \end{aligned}$$

Now, for our energy functional, $I[u] = \int_U L(Du, u, x) dx$, the Euler-Lagrange equation is of the form

$$-\operatorname{div}(L_p) + L_z = 0$$

So, we want $L_p = DF(Du)$ and $L_z = -f$. This implies $L = F(p) - f(x)z$. Therefore, the associated energy functional is of the form

$$I[u] = \int_U \left(F(Du) - f(x)u(x) \right) dx$$

To apply Calculus of Variation, we have to check some questions. Suppose such F exists and satisfies the usual growth and convexity assumptions. We can then see explicitly how the direct method applies. we consider over the space $H_0^1(U)$ for this. We consider the growth condition on $F : \alpha|p|^2 - \beta \leq F(p) \leq C(1 + |p|^2)$ We also consider the convexity of F .

(i) coercivity of I :

$$\begin{aligned} I[u] &= \underbrace{\int_U F(Du) dx}_{\geq \alpha \int_U |Du|^2 dx - \beta|U|} - \underbrace{\int_U f u dx}_{\leq \|f\|_2 \|u\|_2 \leq C\|f\|_2 \|Du\|_2} \\ \implies I[u] &\geq \alpha \|Du\|_2^2 - C_1 \|Du\|_2 - C_2 \end{aligned}$$

This implies I is coercive.

Let $m = \inf_{u \in \mathcal{A}} I[u]$ Choose minimising sequence $\{u_k\}_{k=1}^\infty \subset \mathcal{A}$ such that

$$I[u_k] \rightarrow m$$

Since I is coercive, $\{Du_k\}$ is bounded.

And by Poincaré inequality, $\{u_k\}$ is bounded. So, $u_k \rightharpoonup u$ weakly in $H_0^1(U)$. Since $H_0^1(U)$ is closed, linear subspace of $W^{1,2}(U)$, it is weakly closed, $u \in H_0^1(U)$ Therefore, $u \in \mathcal{A}$.

Claim: $I[u] \leq \liminf_{k \rightarrow \infty} I[u_k]$

By compactness theorem, $u_k \rightarrow u$ strongly in $L^2(U)$ so $\int_U f u_k dx \rightarrow \int_U f u dx$

By convexity of F ,

$$\begin{aligned} \int_U F(Du_k) dx &\geq \int_U F(Du) + \underbrace{\int_U DF(Du) \cdot (Du_k - Du) dx}_{\rightarrow 0 \text{ since } Du_k \rightarrow Du} \\ &\Rightarrow \liminf_{k \rightarrow \infty} \int_U F(Du_k) dx \geq \int_U F(Du) \\ &\Rightarrow \liminf_{k \rightarrow \infty} I[u_k] = \liminf_{k \rightarrow \infty} \left[\int_U F(Du_k) dx - \int_U f u_k dx \right] \\ &\geq \int_U F(Du) dx - \int_U f u dx = I[u] \end{aligned}$$

But $m \leq I[u]$. Hence, $I[u] = m$ and u is a minimizer.

However, if there is no such potential F , we cannot use the variational methods above. We therefore need a different way of constructing solutions to 2.1. This raises the question of what conditions we should impose directly on the nonlinear vector field $\mathbf{a}$.

To motivate the condition we will use, suppose again that $\mathbf{a} = DF$ and that F is convex, then for each $p, q \in \mathbb{R}^n$

$$\begin{aligned} (\mathbf{a}(p) - \mathbf{a}(q)) \cdot (p - q) &= \sum_{i=1}^n (F_{p_i}(p) - F_{p_i}(q))(p_i - q_i) \\ &\int_0^1 \sum_{i,j=1}^n F_{p_i p_j}(p + t(q - p))(p_j - q_j)(p_i - q_i) dt \geq 0, \end{aligned}$$

the last inequality following from the convexity of F .

Definition 2.1.1. A vector field $\mathbf{a} : \mathbb{R}^n \rightarrow \mathbb{R}^n$ is called monotone provided

$$(\mathbf{a}(p) - \mathbf{a}(q)) \cdot (p - q) \geq 0, \quad \forall p, q \in \mathbb{R}^n. \quad (2.3)$$

Under the primary assumption that the non linearity be monotone, we will show below that the quasilinear PDE does indeed possess a weak solution. We henceforth assume that the smooth vector field $\mathbf{a}$ is monotone and satisfies the growth condition

$$|\mathbf{a}(p)| \leq C(1 + |p|) \quad (2.4)$$

We also assume

$$\mathbf{a}(p) \cdot p \geq \alpha |p|^2 - \beta \quad (2.5)$$

for all $p \in \mathbb{R}^n$ and appropriate constants $C, \alpha > 0, \beta \geq 0$.

We will see that 2.5 provides the coercivity needed in the argument. We now construct a solution of the boundary value problem 2.1 by considering a sequence of finite-dimensional approximations.

Let $\{w_k\}_{k=1}^\infty$ be smooth functions which form an orthonormal basis of $H_0^1(U)$ with respect to the inner product

$$(u, v) = \int_U Du \cdot Dv \, dx.$$

(We could for instance take $\{w_k\}_{k=1}^\infty$ to be the set of appropriately normalized eigenfunctions for $-\Delta$ in $H_0^1(U)$.)

We will look for an approximate solution $u_m \in H_0^1(U)$ of the form

$$u_m = \sum_{k=1}^m d_m^k w_k, \quad (2.6)$$

where the coefficients $d_m^1, \dots, d_m^m$ are chosen so that

$$\int_U \mathbf{a}(Du_m) \cdot Dw_k \, dx = \int_U f w_k \, dx, \quad k \in \{1, \dots, m\} \quad (2.7)$$

This amounts to our requiring that u_m solves the "projection" of the problem 2.1 onto the finite-dimensional subspace spanned by $\{w_k\}_{k=1}^m$.

Lemma 2.1.2 (Zeroes of a vector field.).

Assume the continuous function $\mathbf{v} : \mathbb{R}^n \rightarrow \mathbb{R}^n$ satisfies

$$\mathbf{v}(x) \cdot x \geq 0, \quad \text{if } |x| = r, \quad (2.8)$$

for some $r > 0$. Then there exists a point $x \in B(0, r)$ such that

$$\mathbf{v}(x) = 0$$

. Suppose the assertion were false, then $\mathbf{v}(x) \neq 0$ for all $x \in B(0, r)$. Define the contin-

uous mapping $\mathbf{w} : B(0, r) \rightarrow \partial B(0, r)$ by setting

$$\mathbf{w}(x) = -\frac{r}{|\mathbf{v}(x)|} \mathbf{v}(x), \quad x \in B(0, r).$$

According to Brouwer's fixed point theorem, there exists a point $z \in B(0, r)$ with

$$\mathbf{w}(z) = z \tag{2.9}$$

But then $z \in \partial B(0, r)$, and so 2.8 and 2.9 imply the contradiction

$$r^2 = z \cdot z = \mathbf{w}(z) \cdot z = -\frac{r}{|\mathbf{v}(z)|} \mathbf{v}(z) \cdot z \leq 0$$

□

Theorem 2.1.3 (Construction of approximate solutions).

For each integer $m = 1, \dots$, there exists a function u_m of the form 2.6 satisfying the identities 2.7.

Proof. Define the continuous function $\mathbf{v} : \mathbb{R}^m \rightarrow \mathbb{R}^m$, $\mathbf{v} = (v^1, \dots, v^m)$, by setting

$$v^k(d) = \int_U \mathbf{a} \left(\sum_{j=1}^m d_j Dw_j \right) \cdot Dw_k - f w_k \, dx, \text{quad } k \in \{1, \dots, m\} \tag{2.10}$$

for each point $d = (d_1, \dots, d_m) \in \mathbb{R}^m$. Now

$$\begin{aligned} \mathbf{v}(d) \cdot d &= \int_U \mathbf{a} \left(\sum_{j=1}^m d_j Dw_j \right) \cdot \left(\sum_{j=1}^m d_j Dw_j \right) - f \left(\sum_{j=1}^m d_j w_j \right) \, dx \\ &\geq \int_U \alpha \left| \left(\sum_{j=1}^m d_j Dw_j \right) \right|^2 - \beta - f \left(\sum_{j=1}^m d_j w_j \right) \, dx \text{ by 2.5} \\ &= \alpha |d|^2 - \beta |U| - \sum_{j=1}^m \int_U f w_j \, dx \\ &\geq \frac{\alpha}{2} |d|^2 - C \end{aligned}$$

Hence $\mathbf{v}(d) \cdot d \geq 0$ if $|d| = r$, provided we select $r > 0$ sufficiently large. □

We want to take the limit as $m \rightarrow \infty$, and for this will require some uniform esti-

mates.

Theorem 2.1.4 (Energy estimates).

There exists a constant C , depending only on U and $\mathbf{a}$, such that

$$\|u_m\|_{H_0^1(U)} \leq C(1 + \|f\|_{L^2(U)}), \quad m = 1, 2, \dots \quad (2.11)$$

Proof. We have our equation 2.7

$$\int_U \mathbf{a}(Du_m) \cdot Dw_k \, dx = \int_U f w_k \, dx, \quad k \in \{1, \dots, m\}$$

Multiplying d_m^k on both sides

$$d_m^k \int_U \mathbf{a}(Du_m) \cdot Dw_k \, dx = d_m^k \int_U f w_k \, dx, \quad k \in \{1, \dots, m\}$$

we know

$$\begin{aligned} u_m &= \sum_{k=1}^m d_m^k w_k \\ Du_m &= \sum_{k=1}^m d_m^k Dw_k \end{aligned}$$

So we get

$$\begin{aligned}
& \int_U \mathbf{a}(Du_m) \cdot Du_m \, dx = \int_U f u_m \, dx, \\
& \implies \int_U \alpha |Du_m|^2 - \beta \, dx \stackrel{2.5}{\leq} \int_U \mathbf{a}(Du_m) \cdot Du_m \, dx = \int_U f u_m \, dx \\
& \implies \alpha \int_U |Du_m|^2 \, dx \leq \underbrace{C}_{\beta|U|} + \int_U f u_m \, dx \stackrel{\text{Cauchy-ineq-with-}\epsilon}{\leq} C + \epsilon \int_U u_m^2 \, dx + \frac{1}{4\epsilon} \int_U f^2 \, dx \\
& \implies \alpha \int_U |Du_m|^2 \, dx \stackrel{\text{Poincaré's-ineq}}{\leq} C + \epsilon C' \int_U |Du_m|^2 \, dx + \frac{1}{4\epsilon} \|f\|_{L^2(U)}^2 \\
& \implies \alpha \int_U |Du_m|^2 \, dx \leq C'' + \epsilon C'' \int_U |Du_m|^2 \, dx + \frac{1}{4\epsilon} \|f\|_{L^2(U)}^2 \\
& \quad \text{let } \epsilon C'' \leq \alpha/2 \\
& \implies \alpha/2 \int_U |Du_m|^2 \, dx \leq C''(1 + \|f\|_{L^2(U)}^2) \\
& \implies \|Du_m\|_{L^2(U)}^2 \leq C''(1 + \|f\|_{L^2(U)}^2) \\
& \implies \|u_m\|_{H_0^1(U)} \leq C''(1 + \|f\|_{L^2(U)})
\end{aligned}$$

□

We now want to pass to the limit as $m \rightarrow \infty$ and show that the limit is a weak solution of 2.1, i.e., we want to find $u \in H_0^1(U)$ such that

$$\int_U \mathbf{a}(Du) \cdot Dv \, dx = \int_U f v \, dx, \quad \forall v \in H_0^1(U). \quad (2.12)$$

From the estimate 2.11, we can extract a subsequence $\{u_{m_j}\}$ such that

$$u_{m_j} \rightharpoonup u \quad \text{weakly in } H_0^1(U),$$

and

$$u_{m_j} \rightarrow u \quad \text{strongly in } L^2(U)$$

The problem is that weak convergence of Du_{m_j} does not, in general, imply weak convergence of $\mathbf{a}(Du_{m_j})$ to $\mathbf{a}(Du)$. Nonlinear functions are generally not continuous with respect to weak convergence.

Theorem 2.1.5 (Existence of weak solution).

There exists a weak solution of the nonlinear boundary value problem 2.1.

Proof. In the previous theorem, we proved that $\{u_m\}_{m=1}^\infty$ is bounded. Therefore, we can find a subsequence $\{u_{m_j}\}_{j=1}^\infty \subset \{u_m\}_{m=1}^\infty$ and a function $u \in H_0^1(U)$ such that

$$u_{m_j} \rightharpoonup u, \quad \text{weakly in } H_0^1(U) \quad (2.13)$$

and by compactness theorem,

$$u_{m_j} \rightarrow u, \quad \text{weakly in } L^2(U). \quad (2.14)$$

Claim: u satisfies 2.12

Equation 2.4 gives

$$|\mathbf{a}(Du_m)| \leq C(1 + |Du_m|)$$

Therefore, $\{\mathbf{a}(Du_m)\}_{m=1}^\infty$ is bounded in $L^2(U; \mathbb{R}^n)$. So, we can find subsequence $\{\mathbf{a}(Du_{m_j})\}_{j=1}^\infty$ and $\xi \in L^2(U; \mathbb{R}^n)$ such that

$$\mathbf{a}(Du_{m_j}) \rightharpoonup \xi \quad \text{weakly in } L^2(U; \mathbb{R}^n) \quad (2.15)$$

Identity 2.7 gives

$$\begin{aligned} \int_U \mathbf{a}(Du_m) \cdot Dw_k \, dx &= \int_U f w_k \, dx \\ \implies \int_U \xi \cdot Dw_k \, dx &= \int_U f w_k \, dx \end{aligned} \quad (2.16)$$

Since $\{w_k\}_{k=1}^\infty$ form an orthonormal basis of $H_0^1(U)$, consider the sequence of functions $v_n = \sum_{k=1}^n c_k w_k \in H_0^1(U)$ such that $v_n \rightarrow v \in H_0^1(U)$. Equation 2.16 implies

$$\begin{aligned} \implies \int_U \xi \cdot Dv_n \, dx &= \int_U f v_n \, dx \\ \implies \int_U \xi \cdot Dv \, dx &= \int_U f v \, dx \end{aligned} \quad (2.17)$$

Monotonicity condition gives

$$\int_U (\mathbf{a}(Du_m) - \mathbf{a}(Dw)) \cdot (Du_m - Dw) \, dx \geq 0 \quad w \in H_0^1(U), m = 1, \dots \quad (2.18)$$

Substituting equation 2.7 into equation 2.18

$$\begin{aligned}
0 &\leq \int_U (\mathbf{a}(Du_m) - \mathbf{a}(Dw)) \cdot (Du_m - Dw) \, dx \\
&= \int_U \mathbf{a}(Du_m) \cdot Du_m - \mathbf{a}(Du_m) \cdot Dw \\
&\quad - \mathbf{a}(Dw) \cdot Du_m + \mathbf{a}(Dw) \cdot Dw \, dx \\
\implies 0 &\leq \int_U fu_m - \mathbf{a}(Du_m) \cdot Dw - \mathbf{a}(Dw) \cdot (Du_m - Dw) \, dx
\end{aligned}$$

re-indexing $m = m_j \rightarrow \infty$, and recalling 2.13 - 2.51,

$$\int_U fu - \xi \cdot Dw - \mathbf{a}(Dw) \cdot (Du - Dw) \, dx \geq 0$$

Set $v = u$, and using 2.17

$$\begin{aligned}
&\int_U \xi \cdot Du - \xi \cdot Dw - \mathbf{a}(Dw) \cdot (Du - Dw) \, dx \geq 0 \\
\implies &\int_U \xi \cdot (Du - Dw) - \mathbf{a}(Dw) \cdot (Du - Dw) \, dx \geq 0 \\
\implies &\int_U (\xi - \mathbf{a}(Dw)) \cdot (Du - Dw) \, dx \geq 0 \quad \forall w \in H_0^1(U) \quad (2.19)
\end{aligned}$$

Fix $v \in H_0^1(U)$ and set $w = u - \lambda v$ for $\lambda > 0$. Equation 2.19 implies

$$\begin{aligned}
&\int_U (\xi - \mathbf{a}(Du - \lambda Dv)) \cdot (Dv) \, dx \geq 0 \\
&\quad \text{let } \lambda \rightarrow 0 \\
&\int_U (\xi - \mathbf{a}(Du)) \cdot (Dv) \, dx \geq 0 \quad \forall v \in H_0^1(U) \quad (2.20)
\end{aligned}$$

Raplacing v by $-v$, equation 2.20 gives

$$\begin{aligned}
&\int_U (\xi - \mathbf{a}(Du)) \cdot D(-v) \, dx \geq 0. \\
\implies &\int_U (\xi - \mathbf{a}(Du)) \cdot Dv \, dx \leq 0.
\end{aligned}$$

Therefore,

$$\int_U (\xi - \mathbf{a}(Du)) \cdot Dv \, dx = 0. \quad (2.21)$$

2.17 and 2.21 imply

$$\begin{aligned} \int_U \xi \cdot Dv &= \int_U \mathbf{a}(Du) \cdot Dv \, dx \\ \implies \int_U f v \, dx &= \int_U \mathbf{a}(Du) \cdot Dv \, dx \end{aligned}$$

Therefore, u is a weak solution of 2.1. □

To prove the uniqueness, we assume that the vector $\mathbf{a}$ satisfies the condition of strict monotonicity:

$$(\mathbf{a}(p) - \mathbf{a}(q)) \cdot (p - q) \geq \theta |p - q|^2, \quad p, q \in \mathbb{R}^n, \text{ constant } \theta > 0. \quad (2.22)$$

Theorem 2.1.6 (Uniqueness of weak solution).

Assume the strict monotonicity property 2.22 holds. Then there exists only one weak solution of 2.1

Proof. Assume that the functions u and $\tilde{u}$ are two weak solutions. Then

$$\begin{aligned} \int_U \mathbf{a}(Du) \cdot Dv \, dx &= \int_U f v \, dx = \int_U \mathbf{a}(D\tilde{u}) \cdot Dv \, dx \\ \implies \int_U [\mathbf{a}(Du) - \mathbf{a}(D\tilde{u})] \cdot Dv \, dx &= 0, \quad v \in H_0^1(U) \end{aligned}$$

Set $v = u - \tilde{u}$. Then

$$\theta \int_U |Du - D\tilde{u}|^2 \, dx \stackrel{2.22}{\leq} \int_U [\mathbf{a}(Du) - \mathbf{a}(D\tilde{u})] \cdot D(u - \tilde{u}) \, dx = 0$$

So, $Du = D\tilde{u}$ a.e. in U .

Let $w = u - \tilde{u} \in H_0^1(U)$. Then $Dw = 0$ a.e.

By Poincaré inequality,

$$\|w\|_{L^2(U)} \leq C \|Dw\|_{L^2(U)} = 0$$

Hence, $u = \tilde{u}$ a.e. in U . □

2.2 Fixed Point Methods

Theorem 2.2.1 (Schauder's Fixed Point Theorem).

Suppose $K \subset X$ is compact and convex, and assume also

$$A : K \rightarrow K$$

is continuous. Then A has a fixed point in K .

Proof. let $\epsilon > 0$ be fixed. Choose finitely many $u_1, u_2, \dots, u_{N_\epsilon} \in K$ such that Since K is compact, we can cover it with finitely many open balls $\{B^0(u_i, \epsilon)\}_{i=1}^{N_\epsilon}$ i.e.

$$K \subset \bigcup_{i=1}^{N_\epsilon} B^0(u_i, \epsilon) \quad (2.23)$$

, where $u_1, u_2, \dots, u_{N_\epsilon} \in K$. Let K_ϵ be the closed convex hull of the points $\{u_1, u_2, \dots, u_{N_\epsilon}\}$:

$$K_\epsilon = \left\{ \sum_{i=1}^{N_\epsilon} \lambda_i u_i \mid 0 \leq \lambda_i \leq 1, \text{ and } \sum_{i=1}^{N_\epsilon} \lambda_i = 1 \right\}$$

Since K is convex, we note that $K_\epsilon \subset K$. Now, define $P_\epsilon : K \rightarrow K_\epsilon$ by

$$P_\epsilon[u] = \frac{\sum_{i=1}^{N_\epsilon} \text{dist}(u, K - B^0(u_i, \epsilon)) u_i}{\sum_{i=1}^{N_\epsilon} \text{dist}(u, K - B^0(u_i, \epsilon))}, \quad u \in K$$

The denominator is never zero, because of 2.23. Now, clearly P_ϵ is continuous.

Now,

$$\begin{aligned} P_\epsilon[u] - u &= P_\epsilon[u] - \frac{\sum_{i=1}^{N_\epsilon} \text{dist}(u, K - B^0(u_i, \epsilon)) u}{\sum_{i=1}^{N_\epsilon} \text{dist}(u, K - B^0(u_i, \epsilon))} \\ \implies \|P_\epsilon[u] - u\| &= \frac{\sum_{i=1}^{N_\epsilon} \text{dist}(u, K - B^0(u_i, \epsilon)) \|u_i - u\|}{\sum_{i=1}^{N_\epsilon} \text{dist}(u, K - B^0(u_i, \epsilon))} \\ &\stackrel{2.23}{\leq} \epsilon \frac{\sum_{i=1}^{N_\epsilon} \text{dist}(u, K - B^0(u_i, \epsilon))}{\sum_{i=1}^{N_\epsilon} \text{dist}(u, K - B^0(u_i, \epsilon))} = \epsilon. \end{aligned} \quad (2.24)$$

Now, we consider the operator $A_\epsilon : K_\epsilon \rightarrow K_\epsilon$ define by

$$A_\epsilon[u] = P_\epsilon[A[u]], \quad u \in K_\epsilon$$

Now, K_ϵ is homeomorphic to the closed unit ball in $\mathbb{R}^{M_\epsilon}$, for some $M_\epsilon \leq N_\epsilon$.

Brouwer's Fixed Point Theorem implies that there exists a point $u_\epsilon \in K_\epsilon$ with

$$A_\epsilon[u_\epsilon] = u_\epsilon$$

Since K is compact, every sequence has a convergent subsequence. Therefore,

$$u_{\epsilon_j} \rightarrow u \text{ in } X.$$

Claim: u is a fixed point of A .

$$\begin{aligned} \|u_{\epsilon_j} - A[u_{\epsilon_j}]\| &= \|A_{\epsilon_j}[u_{\epsilon_j}] - A[u_{\epsilon_j}]\| \\ &= \|P_{\epsilon_j}[A[u_{\epsilon_j}]] - A[u_{\epsilon_j}]\| \stackrel{2.24}{\leq} \epsilon_j \end{aligned}$$

Since A is continuous, we have $u = A[u]$. □

Definition 2.2.2. A nonlinear mapping $A : K \rightarrow K$ is called compact provided for each bounded sequence $\{u_k\}_{k=1}^\infty$ the sequence $\{A[u_k]\}_{k=1}^\infty$ is precompact; that is, there exists a subsequence $\{u_{k_j}\}_{j=1}^\infty$ such that $\{A[u_{k_j}]\}_{j=1}^\infty$ converges in A .

Theorem 2.2.3 (Schaefer's Fixed Point Theorem.).

Suppose

$$A : X \rightarrow X$$

is continuous and compact mapping. Assume further that the set

$$\{u \in X \mid u = \lambda A[u] \text{ for some } 0 \leq \lambda \leq 1\}$$

is bounded. Then A has a fixed point.

Proof. Choose a constant M sufficiently large that

$$\|u\| < M \text{ if } u = \lambda A[u] \text{ for some } 0 \leq \lambda \leq 1 \tag{2.25}$$

Define

$$\tilde{A}[u] = \begin{cases} A[u] & \text{if } \|A[u]\| \leq M \\ \frac{MA[u]}{\|A[u]\|} & \text{if } \|A[u]\| \geq M \end{cases} \quad (2.26)$$

$\|\tilde{A}[u]\| \leq M$. Observe that $\tilde{A} : B(0, M) \rightarrow B(0, M)$.

Set $K = \text{closed convex hull of } \tilde{A}(B(0, M)) \text{ i.e. } K = \overline{\text{conv}(\tilde{A}(B(0, M)))}$. Hence, $K \subset \tilde{A}(B(0, M))$

Why is $\tilde{A}$ compact map?

Since A is a compact mapping, then for bounded sequence $\{u_k\}$, $A[u_{k_j}] \rightarrow u$ (say) .

Then, since

$$G(x) = \begin{cases} x & \text{if } \|x\| \leq M \\ \frac{Mx}{\|x\|} & \text{if } \|x\| \geq M \end{cases}$$

is continuous, and $\tilde{A}[u] = G(A[u])$, therefore, $\tilde{A}$ is a compact map, and $\tilde{A}(B(0, M))$ is relatively compact, i.e. $\tilde{A}(B(0, M))$ is compact.

Let $C = \tilde{A}(B(0, M))$

Claim:

$$\overline{\text{conv}(C)} = \overline{\text{conv}(\bar{C})}$$

$$C \subset \bar{C}$$

$$\implies \text{conv}(C) \subset \text{conv}(\bar{C})$$

$$\implies \overline{\text{conv}(C)} \subset \overline{\text{conv}(\bar{C})}$$

by the definition of convex hull,

$$C \subset \text{conv}(C)$$

$$\implies \bar{C} \subset \overline{\text{conv}(C)}$$

$$\implies \text{conv}(\bar{C}) \subset \overline{\text{conv}(C)}$$

$$\implies \overline{\text{conv}(\bar{C})} \subset \overline{\text{conv}(C)}$$

Hence, $\overline{\text{conv}(C)} = \overline{\text{conv}(\bar{C})}$.

$\overline{\text{conv}(\bar{C})}$ is a closed subspace of Banach space X . Therefore, $\overline{\text{conv}(\bar{C})}$ is complete.

Since $\bar{C}$ is compact, we can cover it with finitely many $u_1, \dots, u_n$ such that

$$\begin{aligned}\bar{C} &\subset \bigcup_{i=1}^n B(u_i, \epsilon/2) \\ &= \{u_1, \dots, u_n\} + B(0, \epsilon/2) \\ &\subset \text{conv}(\{u_1, \dots, u_n\}) + B(0, \epsilon/2) \\ \text{sum of two convex sets is convex} &\implies \text{conv}(\bar{C}) \subset \text{conv}(\{u_1, \dots, u_n\}) + B(0, \epsilon/2)\end{aligned}$$

Since $\text{conv}(\{u_1, \dots, u_n\})$ is bounded and finite dimensional, $\text{conv}(\{u_1, \dots, u_n\})$ is totally bounded. Hence, it has a finite covering by balls of radius $\epsilon/2$

$$\text{conv}(\{u_1, \dots, u_n\}) \subset \{w_1, \dots, w_m\} + B(0, \epsilon/2)$$

This implies

$$\begin{aligned}\text{conv}(\bar{C}) &\subset \{w_1, \dots, w_m\} + B(0, \epsilon/2) + B(0, \epsilon/2) \\ &= \{w_1, \dots, w_m\} + B(0, \epsilon) \\ &= \bigcup_{i=1}^m B(w_i, \epsilon)\end{aligned}$$

Hence, $\text{conv}(\bar{C})$ is totally bounded, and so is $\overline{\text{conv}(\bar{C})}$. And, $\overline{\text{conv}(\bar{C})} = \overline{\text{conv}(\bar{C})}$ is compact.

K is a compact, convex subset of X . Furthermore $\tilde{A} : K \rightarrow K$.

Schauder's Fixed Point Theorem implies that u is a fixed point of A with

$$\tilde{A}[u] = u. \tag{2.27}$$

Claim: u is a fixed point of A . Suppose u is not fixed point of A . 2.26 and 2.27 imply that

$$\tilde{A}[u] = u \begin{cases} A[u] & \text{if } \|A[u]\| \leq M \\ \frac{MA[u]}{\|A[u]\|} & \text{if } \|A[u]\| \geq M \end{cases}$$

But, whenever $\|A[u]\| \leq M$, $A[u] = u$, which contradicts our assumption.

So, we must have

$$\|A[u]\| \geq M$$

and

$$u = \lambda A[u] \text{ for } \lambda = \frac{M}{\|A[u]\|} < 1 \quad (2.28)$$

But $\|u\| = \|\tilde{A}[u]\| = M$, a contradiction to 2.25 and 2.28.

For further details on the compactness of closed convex hulls, see Mazur's theorem [1, Theorem 11.30, p. 118]. $\square$

Application of the Fixed Point Theorems:

Consider the BVP

$$\begin{cases} -\Delta u + b(Du) + \mu u = 0 & \text{in } U \\ u = 0 & \text{on } \partial U \end{cases} \quad (2.29)$$

where U is bounded with smooth boundary. We also assume $b : \mathbb{R}^n \rightarrow \mathbb{R}$ is smooth, Lipschitz continuous and thus satisfies the growth condition

$$|b(p)| \leq C(|p| + 1) \quad (2.30)$$

for some constant C and all $p \in \mathbb{R}^n$.

Theorem 2.2.4 (Existence).

If $\mu > 0$ is sufficiently large, there exists a function $u \in H^2(U) \cap H_0^1(U)$ solving the boundary-value problem 2.29.

Proof. Given $u \in H_0^1(U)$, set

$$f = -b(Du) \quad (2.31)$$

equation 2.30 implies that $f \in L^2(U)$.

Let $w \in H_0^1(U)$ be the unique weak solution of the linear problem

$$\begin{cases} -\Delta w + \mu w = f & \text{in } U \\ w = 0 & \text{on } \partial U \end{cases} \quad (2.32)$$

Since $f \in L^2(U)$, by the regularity theory, we have $w \in H^2(U)$ satisfying

$$\|w\|_{H^2(U)} \leq C \|f\|_{L^2(U)} \quad (2.33)$$

for some constant C .

Let us write $A[u] = w$ whenever w is derived from u via 2.31 and 2.32. In light of 2.30 and 2.33, we have the estimate

$$\|A[u]\|_{H^2(U)} \leq C(\|u\|_{H_0^1(U)} + 1) \quad (2.34)$$

Claim: $A : H_0^1(U) \rightarrow H_0^1(U)$ is continuous.

If

$$u_k \rightarrow u \quad \text{in } H_0^1(U) \quad (2.35)$$

then, the sequence u_k is bounded and in the view of estimate 2.34 we have

$$\begin{aligned} \|w_k\|_{H^2(U)} &= \|A[u_k]\|_{H^2(U)} < C \\ \implies \sup_k \|w_k\|_{H^2(U)} &< \infty \end{aligned} \quad (2.36)$$

So $\{w_k\}_{k=1}^\infty$ is bounded in $H^2(U)$. And from compactness theorem, the imbedding from $H^2(U) \hookrightarrow H_0^1(U)$ is compact. Therefore there exists a subsequence $\{w_{k_j}\}_{j=1}^\infty$ and a function $w \in H_0^1(U)$ with

$$w_{k_j} \rightarrow \tilde{w} \quad \text{in } H_0^1(U) \quad (2.37)$$

From 2.30,

$$\begin{aligned} |b(Du_k) - b(Du)| &\leq C|Du_k - Du| \\ \implies \|b(Du_k) - b(Du)\|_{L^2(U)} &\leq C\|Du_k - Du\|_{L^2} \rightarrow 0. \end{aligned}$$

Therefore,

$$b(Du_k) \rightarrow b(Du) \quad \text{in } L^2(U)$$

Now, weak form of 2.32 is

$$\int_U Dw_{k_j} \cdot Dv + \mu w_{k_j} v \, dx = - \int_U b(Du_{k_j}) v \, dx$$

for each $v \in H_0^1(U)$. consequently using 2.35 and 2.37, we see

$$\int_U D\tilde{w} \cdot Dv + \mu \tilde{w} v \, dx = - \int_U b(Du) v \, dx$$

for each $v \in H_0^1(U)$. Thus $\tilde{w}$ is the weak solution of 2.32. But $A[u]$ is the unique

solution, therefore, $\tilde{w} = A[u]$.

So far, we have just shown that the subsequence of $\{w_k\}$ converges to $A[u]$. To prove $w_k \rightarrow A[u]$ in $H_0^1(U)$, we need to show that every subsequence of $\{w_k\}$ has a further subsequence converging to $A[u]$. Take an arbitrary subsequence $\{w_{k_j}\}$. Since it is bounded in $H^2(U)$, compactness theorem implies there exists a sub-subsequence $\{w_{k_{j_l}}\}$ and some $w \in H_0^1(U)$ such that

$$w_{k_{j_l}} \rightarrow w \quad \text{in } H_0^1(U).$$

For each $v \in H_0^1(U)$, the weak formulation for $w_{k_{j_l}} = A[u_{k_{j_l}}]$ gives

$$\int_U Dw_{k_{j_l}} \cdot Dv + \mu w_{k_{j_l}} v \, dx = - \int_U b(Du_{k_{j_l}})v \, dx$$

Passing to the limit, we get

$$\int_U Dw \cdot Dv + \mu wv \, dx = - \int_U b(Du)v \, dx \quad \forall v \in H_0^1(U).$$

By uniqueness of the weak solution to the linear PDE, we conclude that $w = A[u]$.

Thus every subsequence of $\{w_k\}$ has a sub-subsequence converging to $A[u]$, which implies the whole sequence $w_k \rightarrow A[u]$ in $H_0^1(U)$. Hence A is continuous. $\square$

Therefore A is continuous.

Compactness of A : Let $\{u_k\}_{k=1}^\infty \subset H_0^1(U)$ be any arbitrary bounded sequence.

Claim: $A[u_k]$ has a convergent subsequence in $H_0^1(U)$. Set $w_k = A[u_k]$. From 2.34 $\{w_k\}$ is uniformly bounded in $H^2(U)$. By compactness theorem, there is a subsequence w_{k_j} and some $w \in H_0^1(U)$ such that

$$w_{k_j} = A[u_{k_j}] \rightarrow w \quad \text{in } H_0^1(U).$$

Therefore, A is compact.

Finally, we must show that if μ is large enough, the set

$$\{u \in H_0^1(U) \mid u = \lambda A[u] \text{ for some } 0 \leq \lambda \leq 1\}$$

is bounded in $H_0^1(U)$. So assume $u \in H_0^1(U)$,

$$u = \lambda A[u] \quad \text{for some } 0 < \lambda \leq 1.$$

Then $\frac{u}{\lambda} = A[u]$ or in other words, $u \in H^2(U) \cap H_0^1(U)$ and

$$-\Delta A[u] + \mu A[u] = -b(Du)$$

or

$$-\Delta u + \mu u = -\lambda b(Du) \quad \text{a.e. in } U.$$

Multiply this identity by u and integrate over U , to compute

$$\begin{aligned} & \int_U (-\Delta u)u + \mu u^2 = \int_U -\lambda b(Du)u \\ \implies & -\sum_{i=1}^n \int_U u_{x_i x_i} u \, dx + \mu u^2 = \int_U -\lambda b(Du)u \\ \implies & \int_U u_{x_i} u_{x_i} \, dx - \underbrace{\int_{\partial U} u_{x_i} u \nu^i \, dS}_{=0} + \mu u^2 = \int_U -\lambda b(Du)u \\ \implies & \int_U |Du|^2 + \mu |u|^2 \, dx = - \int_U \lambda b(Du)u \, dx \leq \int_U C(|Du| + 1)|u| \, dx \\ \implies & \int_U |Du|^2 + \mu |u|^2 \, dx \leq C \int_U |Du||u| \, dx + C \int_U |u| \, dx \\ & \stackrel{\text{Young's-ineq}}{\leq} \frac{1}{2} \int_U |Du|^2 \, dx + C \int_U |u|^2 \, dx + C \int_U \frac{1}{2}|u|^2 + \frac{1}{2} \, dx \\ & \leq \frac{1}{2} \int_U |Du|^2 \, dx + C \int_U |u|^2 + 1 \, dx \\ \implies & \int_U \frac{1}{2}|Du|^2 + (\mu - C)|u|^2 \, dx \leq C \end{aligned}$$

Thus if $\mu > 0$ is sufficiently large, we have $\|u\|_{H_0^1(U)} \leq C$, for some constant C that does not depend on $0 \leq \lambda \leq 1$.

Applying Schaefer's Fixed Point Theorem in the space $X = H_0^1(U)$, we conclude that A has a fixed point $u \in H_0^1(U) \cap H^2(U)$, which in turn solves 2.29. $\square$

2.3 Method of Subsolutions and Supersolutions

In this section, we focus on another method to prove the existence of solution of elliptic PDEs. But first of all, let's consider this boundary value problem for the nonlinear Poisson equation:

$$\begin{cases} -\Delta u = f(u) & \text{in } U, \\ u = 0 & \text{on } \partial U. \end{cases} \quad (2.38)$$

where $f : \mathbb{R} \rightarrow \mathbb{R}$ is smooth, with

$$|f'| \leq C, \quad z \in \mathbb{R} \quad (2.39)$$

for some constant C .

We will show that if we can find a subsolution $\underline{u}$ and a supersolution $\bar{u}$ of the boundary value problem 2.38 and if $\underline{u} \leq \bar{u}$, then there exists a solution u satisfying

$$\underline{u} \leq u \leq \bar{u}.$$

Definition 2.3.1.

- We say that $\bar{u} \in H^1(U)$ is a weak supersolution of problem 2.38 if

$$\int_U D\bar{u} \cdot Dv \, dx \geq \int_U f(\bar{u})v \, dx \quad (2.40)$$

for each $v \in H_0^1(U)$, $v \geq 0$ a.e.

- Similarly, $\underline{u} \in H^1(U)$ is a weak subsolution provided

$$\int_U D\underline{u} \cdot Dv \, dx \leq \int_U f(\underline{u})v \, dx \quad (2.41)$$

for each $v \in H_0^1(U)$, $v \geq 0$ a.e.

- We say $u \in H_0^1(U)$ is a weak solution of 2.38 if

$$\int_U Du \cdot Dv \, dx = \int_U f(u)v \, dx$$

for each $v \in H_0^1(U)$

Remark 2.3.2. If $\bar{u}, \underline{u} \in C^2(U)$, then from 2.40 and 2.41 it follows that

$$-\Delta \bar{u} \geq f(\bar{u}), \quad \Delta \underline{u} \leq f(\underline{u}), \quad \text{in } U.$$

Theorem 2.3.3 (Existence of a solution between sub- and supersolutions).

Assume there exist a weak supersolution $\bar{u}$ and a weak subsolution $\underline{u}$ of 2.38, satisfying

$$\underline{u}_0, \bar{u} \geq \text{ on } \partial U \text{ in the trace sense, } \underline{u} \leq \bar{u} \text{ a.e. in } U. \quad (2.42)$$

Then there exists a weak solution u of 2.38, such that

$$\underline{u} \leq u \leq \bar{u} \quad \text{a.e. in } U.$$

idea of the proof: The main idea is to construct a sequence of linear problems, starting with the subsolution $u_0 = \underline{u}$. We then choose a number $\lambda > 0$ so that $f(z) + \lambda z$ is nondecreasing and use it to define inductively u_{k+1} from u_k . We then show that $\underline{u} = u_0 \leq u_1 \leq \dots \leq u_k \leq \bar{u}$. Thus u_k converges to some u lying between the subsolution and supersolution. Then finally, we pass to the limit in the weak formulation to show that u is a weak solution.

Proof. Fix $\lambda > 0$ so large that

$$z \mapsto f(z) + \lambda z \text{ is non decreasing.} \quad (2.43)$$

$$f'(z) + \lambda > 0, \quad \lambda > -C$$

let $u_0 = \underline{u}$.

for functions u_k , $k = 0, 1, 2, \dots$, we define $u_{k+1} \in H_0^1(U)$, inductively, to be the unique weak solution of the linear BVP

$$\begin{cases} \Delta u_{k+1} + \lambda u_{k+1} = f(u_k) + \lambda u_k & \text{in } U, \\ u_{k+1} = 0 & \text{on } \partial U. \end{cases} \quad (2.44)$$

NOTE: Here, u_k is known, so $f(u_k) + \lambda u_k$ is now a known function of x .

Claim:

$$\underline{u} \leq u_0 \leq u_1 \leq \dots \leq u_k \leq \dots \text{ a.e. in } U. \quad (2.45)$$

For $k = 0$, we note that 2.44 gives

$$\Delta u_1 + \lambda u_1 = f(u_0) + \lambda u_0$$

OR

$$\int_U Du_1 \cdot Dv + \lambda u_1 v \, dx = \int_U (f(u_0) + \lambda u_0) v \, dx, \quad v \in H_0^1(U) \quad (2.46)$$

subtracting 2.46 from 2.41

$$\begin{aligned} \int_U Du_0 \cdot Dv \, dx - \left(\int_U Du_1 \cdot Dv + \lambda u_1 v \, dx \right) &\leq \int_U f(u_0) v - \int_U (f(u_0) + \lambda u_0) v \, dx \\ \implies \int_U (Du_0 - Du_1) \cdot Dv \, dx - \lambda \int_U u_1 v \, dx &\leq -\lambda \int_U u_0 v \, dx \\ \implies \int_U (Du_0 - Du_1) \cdot Dv \, dx + \lambda \int_U (u_0 - u_1) v \, dx &\leq 0 \end{aligned}$$

Set $v = (u_0 - u_1)^+ \in H_0^1(U)$, $v \geq 0$ a.e.

We recall that

$$\begin{aligned} (u_0 - u_1)^+ &= \max(u_0 - u_1, 0) \\ D(u_0 - u_1)^+ &= \begin{cases} D(u_0 - u_1) & \text{a.e. on } \{u_0 \geq u_1\} \\ 0 & \text{a.e. on } \{u_0 \leq u_1\} \end{cases} \end{aligned}$$

Therefore,

$$\begin{aligned} \int_U (Du_0 - Du_1) \cdot Dv \, dx + \lambda \int_U (u_0 - u_1) v \, dx &\leq 0 \\ \implies \int_U (Du_0 - Du_1) \cdot D(u_0 - u_1)^+ \, dx + \lambda \int_U (u_0 - u_1)(u_0 - u_1)^+ &\leq 0 \\ \implies \int_{\{u_0 \geq u_1\}} |Du_0 - Du_1|^2 + \lambda(u_0 - u_1)^2 \, dx &\leq 0 \end{aligned}$$

This is impossible since $\lambda > 0$. Therefore $u_0 \leq u_1$ a.e. in U . Now assume inductively that

$$u_{k-1} \leq u_k \quad \text{a.e. in } U. \quad (2.47)$$

equation 2.44 gives

$$\begin{aligned} \int_U Du_{k+1} \cdot Dv + \lambda u_{k+1} v \, dx &= \int_U (f(u_k) + \lambda u_k) v \, dx, \\ &\text{and} \\ \int_U Du_k \cdot Dv + \lambda u_k v \, dx &= \int_U (f(u_{k-1}) + \lambda u_{k-1}) v \, dx, \end{aligned} \quad (2.48)$$

for each $v \in H_0^1(U)$. Subtracting, we get

$$\begin{aligned} &\left(\int_U Du_k \cdot Dv + \lambda u_k v \, dx \right) - \left(\int_U Du_{k+1} \cdot Dv + \lambda u_{k+1} v \, dx \right) \\ &= \left(\int_U (f(u_{k-1}) + \lambda u_{k-1}) v \, dx \right) - \left(\int_U (f(u_k) + \lambda u_k) v \, dx \right) \\ &\implies \int_U \left(Du_k - Du_{k+1} \right) \cdot Dv + \lambda (u_k - u_{k+1}) v \, dx \\ &= \int_U \left((f(u_{k-1}) + \lambda u_{k-1}) - (f(u_k) + \lambda u_k) \right) v \, dx \end{aligned}$$

Set $v = (u_k - u_{k+1})^+ \in H_0^1(U)$. We deduce

$$\begin{aligned} &\int_U \left[(Du_k - Du_{k+1}) \cdot D(u_k - u_{k+1})^+ + \lambda (u_k - u_{k+1})(u_k - u_{k+1})^+ \right] dx \\ &= \int_U \left[(f(u_{k-1}) + \lambda u_{k-1}) - (f(u_k) + \lambda u_k) \right] (u_k - u_{k+1})^+ dx \\ &\implies \int_{\{u_k \geq u_{k+1}\}} \left[|D(u_k - u_{k+1})|^2 + \lambda (u_k - u_{k+1})^2 \right] dx \\ &= \int_U \left[(f(u_{k-1}) + \lambda u_{k-1}) - (f(u_k) + \lambda u_k) \right] (u_k - u_{k+1})^+ dx \\ &\stackrel{2.47}{\leq} 0 \\ &\stackrel{2.43}{\leq} 0 \end{aligned}$$

Therefore, $u_k \leq u_{k+1}$ a.e. in U .

Claim:

$$u_k \leq \bar{u} \quad \text{a.e. in } U, \quad k = 0, 1, 2, \dots \quad (2.49)$$

This is true for $k = 0$ by 2.42. Assume that

$$u_k \leq \bar{u} \quad \text{a.e. in } U. \quad (2.50)$$

Subtracting 2.40 from 2.48

$$\begin{aligned}
& \left(\int_U Du_{k+1} \cdot Dv + \lambda u_{k+1} v \, dx \right) - \left(\int_U D\bar{u} \cdot Dv \, dx \right) \\
& \leq \left(\int_U (f(u_k) + \lambda u_k) v \, dx \right) - \left(\int_U f(\bar{u}) v \, dx \right) \\
& \implies \left(\int_U (D(u_{k+1} - \bar{u})) \cdot v + \lambda(u_{k+1} - \bar{u}) v \, dx \right) \\
& \leq \int_U \left((f(u_k) + \lambda u_k) - (f(\bar{u})) \right) v \, dx
\end{aligned}$$

Subtracting $\int_U \lambda \bar{u} v \, dx$ on both sides

$$\begin{aligned}
& \int_U (D(u_{k+1} - \bar{u})) \cdot v + \lambda(u_{k+1} - \bar{u}) v \, dx \\
& \leq \int_U \left((f(u_k) + \lambda u_k) - (f(\bar{u}) + \lambda \bar{u}) \right) v \, dx
\end{aligned}$$

Set $v = (u_{k+1} - \bar{u})^+$

$$\begin{aligned}
& \int_{\{u_{k+1} \geq \bar{u}\}} |D(u_{k+1} - \bar{u})|^2 + \lambda(u_{k+1} - \bar{u})^2 \, dx \\
& \leq \int_U \left((f(u_k) + \lambda u_k) - (f(\bar{u}) + \lambda \bar{u}) \right) (u_{k+1} - \bar{u})^+ \, dx \\
& \stackrel{2.50}{\leq} 0 \\
& \stackrel{2.43}{\leq} 0
\end{aligned}$$

Therefore, $u_{k+1} \leq \bar{u}$ a.e. in U . 2.45 and 2.49 imply

$$\underline{u} \leq u_0 \leq \dots \leq u_k \leq u_{k+1} \leq \dots \leq \bar{u} \text{ a.e. in } U. \quad (2.51)$$

Therefore

$$u(x) = \lim_{k \rightarrow \infty} u_k(x) \text{ for a.e. } x \text{ since } \{u_k(x)\} \text{ is monotone increasing and bounded.}$$

Claim: $u_k \rightarrow u$ in $L^2(U)$ Define $g(x) = \max\{|\underline{u}|, |\bar{u}|\}$. Since $\underline{u}, \bar{u} \in H^1(U)$, $g \in L^2(U)$. We also note that

$$|u_k - u| \leq |u_k| + |u| \leq 2g$$

since $u_k \rightarrow u$ a.e., Dominated Convergence Theorem gives

$$\int_U |u_k - u|^2 dx \rightarrow 0.$$

Hence $u_k \rightarrow u$ in $L^2(U)$. from eqn 2.39 we have $|f'| \leq C$. By M.V.T,

$$\begin{aligned} |f(z)| &\leq C(1 + |z|) \\ \implies |f(u_k)| &\leq C(1 + |u_k|) \\ \implies \|f(u_k)\|_{L^2(U)} &\leq C \|1 + |u_k|\| \leq C(1 + \|u_k\|_{L^2(U)}) \end{aligned}$$

Weak form of 2.44

$$\begin{aligned} \int_U Du_{k+1} \cdot Dv + \lambda u_{k+1} v dx &= \int_U (f(u_k) + \lambda u_k) v dx \\ \text{Set } v &= u_{k+1} \\ \int_U |Du_{k+1}|^2 + \lambda |u_{k+1}|^2 dx &= \int_U (f(u_k) + \lambda u_k)(u_{k+1}) dx \\ \implies \|Du_{k+1}\|_{L^2(U)}^2 + \lambda \|u_{k+1}\|_{L^2(U)}^2 &\stackrel{\text{Cauchy-Schwartz}}{\leq} \underbrace{\|f(u_k) + \lambda u_k\|_{L^2(U)} \|u_{k+1}\|_{L^2(U)}}_{f(u_k) \text{ and } \{u_k\} \text{ are both bounded in } L^2(U)} \leq C \\ \implies \min\{1, \lambda\}(\|Du_{k+1}\|_{L^2(U)}^2 + \|u_{k+1}\|_{L^2(U)}^2) &\leq C \\ \implies \sup_k \|u_k\|_{H_0^1(U)} &< \infty \end{aligned}$$

Hence there is a subsequence $u_{k_j} \rightharpoonup u$ which converges weakly in $H_0^1(U)$ to $u \in H_0^1(U)$.

Verifying u is a weak solutin of 2.38. from 2.44

$$\begin{aligned} \int_U Du_{k+1} \cdot Dv + \lambda u_{k+1} v dx &= \int_U (f(u_k) + \lambda u_k) v dx \\ \text{letting } k &\rightarrow \infty \\ \implies \int_U Du \cdot Dv + \lambda uv dx &= \int_U (f(u) + \lambda u) v dx \\ \implies \int_U Du \cdot Dv dx &= \int_U f(u) v dx \end{aligned}$$

□

2.4 Nonexistence of Solutions

We now investigate a nonlinear elliptic PDE to prove nonexistence of nontrivial solutions. Consider the boundary value problem

$$\begin{cases} \Delta u = |u|^{p-1}u & \text{in } U \\ u = 0 & \text{on } \partial U. \end{cases} \quad (2.52)$$

Recall from section 1.6.3 that, for

$$1 < p < \frac{n+2}{n-2} \quad (2.53)$$

the Mountain Pass Theorem, can be used to prove the existence of a nontrivial weak solution.

We now consider the same type of problem in

$$\frac{n+2}{n-2} < p < \infty \quad (2.54)$$

We will show, under a suitable geometric condition on U , that the only smooth solution is the trivial solution $u \equiv 0$. We see therefore that the restriction to condition 2.53 that appeared in section 1.6.3 was in some sense natural, and consequently say $p = \frac{n+2}{n-2}$ is a critical exponent.

Definition 2.4.1. An open set U is called star shaped with respect to 0 provided for each $x \in \bar{U}$, the line segment

$$\{\lambda x \mid 0 \leq \lambda \leq 1\}$$

lies in $\bar{U}$.

Clearly if U is convex and $0 \in U$, then U is star shaped with respect to 0. But a general star-shaped region need not be convex.

Lemma 2.4.2 (Normals to a star-shaped region.).

Assume ∂U is C^1 and U is starshaped with respect to 0. Then

$$x \cdot \nu(x) \geq 0, \quad \text{for all } x \in \partial U,$$

where ν denotes the unit outward normal.

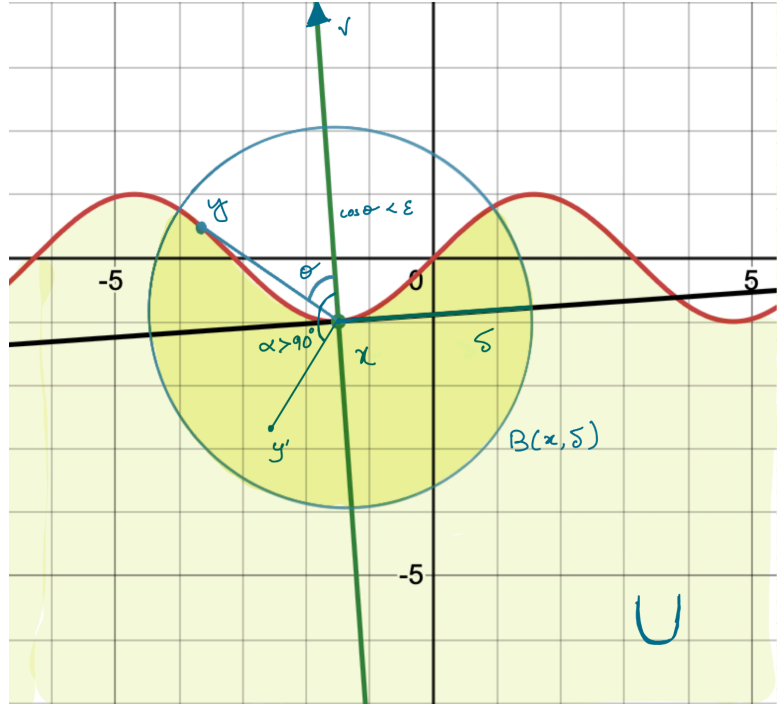


Figure 2.4.1: Boundary of U is C^1 , that means locally it is a C^1 map. In the δ -neighbourhood of x , we can always find points y such that the vector $y - x$ and the unit normal vector ν at x makes an angle such that $\cos\theta < \epsilon$.

. Since ∂U is C^1 , if $x \in \partial U$, then for each $\epsilon > 0$ there exists $\delta > 0$ such that $|y - x| < \delta$ and $y \in \bar{U}$ imply $\nu(x) \cdot \frac{(y-x)}{|y-x|} \leq \epsilon$

In particular

$$\limsup_{\substack{y \rightarrow x \\ y \in \bar{U}}} \nu(x) \cdot \frac{(y-x)}{|y-x|} \leq 0$$

Let $y = \lambda x$ for $0 < \lambda < 1$. Then $y \in \bar{U}$, since U is star shaped. Thus

$$\begin{aligned} \nu(x) \cdot \frac{x}{|x|} &= \lim_{\lambda \rightarrow 1^-} \nu(x) \cdot \frac{(\lambda x - x)}{|\lambda x - x|} = \lim_{\lambda \rightarrow 1^-} \nu(x) \cdot \frac{x(\lambda - 1)}{|x| |\lambda - 1|} \\ &= \lim_{\lambda \rightarrow 1^-} \nu(x) \cdot \frac{x(\lambda - 1)}{|x| (1 - \lambda)} \geq 0. \end{aligned}$$

□

2.4.1 Derrick-Pohozaev identity

Theorem 2.4.3 (Nonexistence of nontrivial solution).

solutions Assume $u \in C^2(\bar{U})$ is a solution of problem 2.52 and the exponent p satisfies inequality 2.54. Suppose further U is a star-shaped with respect to 0 and ∂U is C^1 . Then

$$u \equiv 0 \text{ within } U.$$

Proof. Multiplying the PDE by $x \cdot Du$ and integrating over U ,

$$\underbrace{\int_U (-\Delta u)(x \cdot Du) \, dx}_A = \underbrace{\int_U |u|^{p-1} u (x \cdot Du) \, dx}_B \quad (2.55)$$

$$\begin{aligned} A &= \int_U (\Delta u)(x \cdot Du) \, dx \\ &= \int_U \left(-\sum_{i=1}^n u_{x_i x_i} \right) \left(\sum_{j=1}^n x_j u_{x_j} \right) \\ &= -\sum_{i,j=1}^n \int_U u_{x_i x_i} x_j u_{x_j} \\ &\stackrel{\text{Integrating-by-parts}}{=} -\sum_{i,j=1}^n \left[-\int_U u_{x_i} (x_j u_{x_j})_{x_i} \, dx + \int_{\partial U} u_{x_i} x_j u_{x_j} \nu^i \, dS \right] \\ &= \underbrace{\sum_{i,j=1}^n \int_U u_{x_i} (x_j u_{x_j})_{x_i} \, dx}_{A_1} + \underbrace{\left(-\sum_{i,j=1}^n \int_{\partial U} u_{x_i} x_j u_{x_j} \nu^i \, dS \right)}_{A_2} \quad (2.56) \end{aligned}$$

Now,

$$\begin{aligned}
 A_1 &= \sum_{i,j=1}^n \int_U u_{x_i} (x_j u_{x_j})_{x_i} dx \\
 &= \sum_{i,j=1}^n \int_U u_{x_i} (\delta_{ij} u_{x_j} + x_j u_{x_j x_i}) \\
 &= \int_U |Du|^2 + \sum_{i,j=1}^n u_{x_i} x_j u_{x_j x_i} dx \\
 &= \int_U |Du|^2 + \underbrace{\sum_{j=1}^n \left(\frac{|Du|^2}{2} \right)_{x_j} x_j}_{C} dx \tag{2.57}
 \end{aligned}$$

$$\begin{aligned}
 C &= \sum_{j=1}^n \int_U \left(\frac{|Du|^2}{2} \right)_{x_j} x_j dx \\
 &\stackrel{\text{Integrating-by-parts}}{=} \sum_{j=1}^n \left(- \int_U \frac{|Du|^2}{2} dx + \int_{\partial U} \frac{|Du|^2}{2} x_j \nu^j dS \right) \\
 &= - \sum_{j=1}^n \int_U \frac{|Du|^2}{2} dx + \int_{\partial U} \frac{|Du|^2}{2} \sum_{j=1}^n x_j \nu^j dS \\
 &= - \frac{n}{2} \int_U \frac{|Du|^2}{2} dx + \int_{\partial U} \frac{|Du|^2}{2} x \cdot \nu
 \end{aligned}$$

Substituting this value of C into A_1 :

$$\begin{aligned}
 A_1 &= \int_U |Du|^2 - \frac{n}{2} \int_U \frac{|Du|^2}{2} dx + \int_{\partial U} \frac{|Du|^2}{2} x \cdot \nu \\
 &= \left(1 - \frac{n}{2} \right) \int_U \frac{|Du|^2}{2} dx + \int_{\partial U} \frac{|Du|^2}{2} x \cdot \nu \tag{2.58}
 \end{aligned}$$

Since $u = 0$ on ∂U , we have $Du(x)$ perpendicular to the level surface of $u(x)$, which is

the boundary ∂U . Therefore, $Du(x) \parallel \nu$ at each $x \in \partial U$. Thus, $Du(x) = \pm |Du| \nu$.

$$\begin{aligned}
 A_2 &= \sum_{i,j=1}^n \int_{\partial U} u_{x_i} x_j u_{x_j} \nu^i dS \\
 &= \int_{\partial U} (Du(x) \cdot \nu^i)(x \cdot Du(x)) dS \\
 &= \int_{\partial U} (|Du| |\nu|^2)(x \cdot |Du| \nu) dS \\
 &= - \int_{\partial U} |Du|^2 (\nu \cdot x) dS.
 \end{aligned} \tag{2.59}$$

Combining 2.56 - 2.59

$$A = \left(1 - \frac{n}{2}\right) \int_U \frac{|Du|^2}{2} dx + \int_{\partial U} \frac{|Du|^2}{2} (x \cdot \nu) - \int_{\partial U} |Du|^2 (\nu \cdot x) dS \tag{2.60}$$

$$= \left(1 - \frac{n}{2}\right) \int_U \frac{|Du|^2}{2} dx - \frac{1}{2} \int_{\partial U} |Du|^2 (\nu \cdot x) dS \tag{2.61}$$

from 2.55

$$\begin{aligned}
 B &= \int_U |u|^{p-1} u (x \cdot Du) dx \\
 &= \sum_{j=1}^n \int_U |u|^{p-1} u x_j u_{x_j} dx \\
 &= \sum_{j=1}^n \int_U \underbrace{\left(\frac{|u|^{p+1}}{p+1} \right) x_j}_{D} dx \\
 &\stackrel{\text{Integration-by-parts}}{=} \sum_{j=1}^n \left[- \int_U \frac{|u|^{p+1}}{p+1} dx + \underbrace{\int_{\partial U} \frac{|u|^{p+1}}{p+1} x_j \nu^j dS}_{=0 \text{ as } u=0 \text{ on } \partial U} \right] \\
 &= \frac{-n}{p+1} \int_U |u|^{p+1} dx
 \end{aligned}$$

So,

$$\begin{aligned}
& A = B \\
\Rightarrow & \left(\frac{n-2}{2} \right) \int_U \frac{|Du|^2}{2} dx + \underbrace{\frac{1}{2} \int_{\partial U} |Du|^2 (\nu \cdot x) dS}_{\geq 0 \text{ by lemma 2.4.2}} = \frac{n}{p+1} \int_U |u|^{p+1} dx \\
\Rightarrow & \left(\frac{n-2}{2} \right) \int_U \frac{|Du|^2}{2} dx \leq \frac{n}{p+1} \int_U |u|^{p+1} dx \tag{2.62}
\end{aligned}$$

Now, multiplying 2.52 by u and integrating over U :

$$\begin{aligned}
& \int_U -\Delta u \cdot u dx = \int_U |u|^{p-1} u^2 dx = \int_U |u|^{p+1} dx \\
& \Rightarrow - \sum_{i=1}^n \int_U u_{x_i x_i} u dx = \int_U |u|^{p+1} dx \\
& \Rightarrow \stackrel{Int-by-parts}{=} - \sum_{i=1}^n \left[- \int_U u_{x_i}^2 dx + \underbrace{\int_{\partial U} u_{x_i} u v^i dS}_{=0 \text{ as } u=0 \text{ on } \partial U} \right] = \int_U |u|^{p+1} dx \\
& \Rightarrow \sum_{i=1}^n \int_U u_{x_i}^2 dx = \int_U |u|^{p+1} dx \\
& \Rightarrow \int_U |Du|^2 dx = \int_U |u|^{p+1} dx
\end{aligned}$$

Substituting the above equality into 2.62

$$\left(\frac{n-2}{2} - \frac{n}{p+1} \right) \int_U |u|^{p+1} dx \leq 0$$

If u is the nontrivial solution, then $\int_U |u|^{p+1} dx \geq 0$.

This implies $\frac{n-2}{2} \leq \frac{n}{p+1}$ or $p \leq \frac{n+2}{n-2}$.

□

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